

Equilibrium in a DeFi Lending Market

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Abstract

We develop a model of Decentralized Finance (DeFi) lending platforms that set interest rates as programmable functions of the utilization of available funds. These platforms are unable to incorporate off-chain information into the platform's interest rates, leading to inefficient DeFi equilibria that feature excess demand or supply. Absent uncertainty about withdrawals, these inefficiencies can be made arbitrarily small through an interest rate function that is highly sensitive to changes in utilization. In contrast, in the presence of withdrawal shocks, the optimal interest rate function must balance a trade-off between the efficiency and volatility of interest rates.

Keywords: Decentralized Finance, DeFi, DeFi Lending, Protocol for Loanable Funds, PLF

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1 Introduction

Decentralized finance (DeFi) is a new technology that allows users to access traditional financial services without relying on a trusted intermediary. In this paper, we study a new type of DeFi protocol, termed DeFi lending, that allows agents to borrow and lend funds in a peer-to-peer fashion through smart contracts on a blockchain (e.g., Ethereum). These smart contracts, generally referred to as a protocol for loanable funds (PLF), allow investors to freely supply loanable funds from which other agents may borrow. Agents who borrow these funds pay interest on their loans, and the PLF passes along the borrower interest payments to the fund’s suppliers. To date, the three largest DeFi lending platforms house an aggregate supply of loanable funds of over \$18bn. Yet, under current design, less than half of these funds are actively borrowed, with the remaining funds idled in the platform, raising questions about the economic efficiency of those lending markets and their design.

This paper introduces a new theoretical framework to study a decentralized lending market that operates with a PLF interest rate-setting mechanism. A defining feature of DeFi lending arises from the technical constraints that limit the ability of blockchain applications to incorporate external (off-chain) information in their functioning (see e.g., [Capponi et al. 2023](#) and [John et al. 2023](#)). Due to these constraints, DeFi lending relies on a pre-programmed interest rate function, which sets the borrowing and lending rates strictly as a function of the observed ratio of borrowed-to-available loanable funds, referred to as *the utilization rate*. Given that DeFi lending represents a young and growing market, a natural question arises regarding the way that PLF interest rate functions are set in practice and what key features should drive their design. In order to address these questions, we derive a model of a DeFi lending market that operates under a PLF and demonstrate a key efficiency-volatility trade-off that the PLF interest rate function must balance.

We first focus on the case of a DeFi lending market without uncertainty about the

lenders’ demand for withdrawals. After proving the existence and uniqueness of a DeFi equilibrium for a general class of programmable interest rate functions, we characterize the key informational disadvantage of a PLF interest rate-setting mechanism. Given that a PLF doesn’t have access to off-chain information, it cannot identify the current market conditions in the face of simultaneous shocks to borrowing demand and lending supply. This information disadvantage makes it infeasible to design a PLF function under which the DeFi lending market always clears. As a consequence, the DeFi platform will generically feature excess supply or demand at the equilibrium PLF interest rates, leading to an efficiency loss when compared to the competitive equilibrium.

We then investigate the optimal design of the PLF interest rate function. We show that without uncertainty about the PLF lenders’ demand for withdrawals, the inefficiency of DeFi equilibrium interest rates can be made arbitrarily small under the appropriate design. Importantly, this result holds for any realization of the underlying market conditions and does not require the platform to be able to anticipate those conditions nor to adjust the interest rate function when those conditions change. Rather, such an outcome is achieved by setting the slope of the interest rate function as steep as possible to maximize interest rate sensitivity and minimize deviations from full utilization in all market conditions. This insight is important for the largest DeFi lending platforms (e.g., AAVE and Compound) as these platforms typically rely on low slope parameters, which generates a large under-utilization of the platform’s funds.

Importantly though, we show that this efficiency result can be undermined when lenders receive shocks that force them to withdraw their funds prematurely. In particular, upon receiving a (random) withdrawal shock, lenders withdraw from the platform, which introduces noise in the utilization rate. With a positively-sloped PLF, this noise translates into interest rate risk for the market participants who retain their borrowing (or lending) positions. When agents are risk-averse, the platform faces an additional trade-off between increasing lending

market efficiency and reducing interest rate volatility, yielding an interior solution for the slope of the PLF interest rate function.

A Primer on PLF Interest Rates A DeFi lending market is characterized by a Protocol for Loanable Funds, which serves a similar role to that of a financial intermediary within a traditional lending market. The PLF is a set of smart contracts that encodes transparent rules for setting interest rates, which is this paper’s main object of study. The PLF rules are typically defined by a single pre-determined interest rate function which sets the borrower interest rate directly and the lender interest rate indirectly. Specifically, the borrower interest rate is given by a particular point on the PLF interest rate function, whereas the lender interest rate is the implied rate from passing borrower payments onto lenders.

A PLF sets the borrower interest rate by inputting on-chain information regarding lending supply and borrowing demand into the previously referenced interest rate function. More precisely, the PLF sets its borrowing interest rate as a function of the *utilization rate*, defined as

$$\text{Utilization Rate} = \frac{\text{Observed Borrowing}}{\text{Observed Lending}}. \quad (1.1)$$

In turn, the borrower’s interest rate is determined as

$$\text{Borrower IR} = \rho(\text{Utilization Rate}), \quad (1.2)$$

where ρ denotes the exogenous interest rate function that defines the PLF. The interest rate function, ρ , is specified as an increasing function of the utilization rate so that borrower interest rates increase with observed borrowing demand and decrease with observed lending supply. Then, since the PLF passes borrower payments onto lenders, the PLF lender interest

rate must satisfy the following equation:¹

$$\underbrace{\text{Observed Lending} \cdot \text{Lender IR}}_{\text{Lender Receipts}} = \underbrace{\text{Observed Borrowing} \cdot \text{Borrower IR}}_{\text{Borrower Payments}} \quad (1.3)$$

Combining these features yields the mechanical rule for determining lender interest rates in a PLF as a function of the observed utilization rate:

$$\text{Lender IR} = \text{Utilization Rate} \cdot \rho(\text{Utilization Rate}) \quad (1.4)$$

Finally, we note that given the pseudonymous nature of blockchain accounts, the current form of DeFi lending is collateralized lending and, therefore, analogous to margin lending or repo. As mentioned above, our focus in this paper is on the interest rate-setting mechanism, which is determined independently of the collateral mechanism.

Model Preview We derive a model of DeFi lending to study the interest rate-setting procedure of a PLF. Each period, a mass of both potential borrowers and lenders randomly arrive. Our analysis first considers a baseline case without uncertainty about withdrawal demand. Potential borrowers arrive with no capital but have an investment opportunity that generates heterogeneous returns for each such borrower. Potential lenders arrive with a unit of capital with which they can lend to the platform or invest in an alternative outside option, also with heterogeneous returns for each such lender. Given the design of the PLF, the borrowers whose investment opportunities generate a higher return than the PLF borrowing rate will optimally borrow from the PLF. Similarly, lenders with outside options that generate lower returns than the PLF lending rate will optimally lend to the PLF.² We characterize a

¹In practice, a PLF generally passes a fixed proportion of payments onto the lender. That proportion is generally close to 100%, so we abstract from this detail and assume a 100% pass-through.

²These assumptions are made to generate downward-sloping demand curves and upward-sloping supply curves for loanable funds.

DeFi lending equilibrium as a fixed point at which utilization is consistent with the market participants' optimal lending and borrowing decisions when taking into account that the PLF sets interest rates as a function of utilization. We establish the existence and uniqueness of a DeFi lending equilibrium for a general class of interest rate functions.

We then turn to analyze the performance of the programmable interest rate design in DeFi lending. First, we show that for any fixed PLF interest rate function, the DeFi lending equilibrium generates a lower level of efficiency when compared to a perfectly competitive market-clearing equilibrium. To understand this result, first note that the first-best outcome is achieved when lending supply and borrowing demand is equalized for any underlying state of the credit market (i.e., markets always clear without rationing). Achieving this outcome would require an equilibrium utilization rate of unity in each state of the world. Yet, because the PLF interest rates adjust only to changes in the utilization rate, achieving a utilization rate of unity in every state of the world requires fixing the PLF interest rate to a constant rate given by $\rho(1)$. This fixed interest rate is, however, inconsistent with the market clearing interest rate in each state of the world, as market clearing requires rates to increase (decrease) when there is a positive (negative) shock to borrowing demand or a negative (positive) shock to lending supply. Therefore, a DeFi lending market cannot simultaneously ensure that interest rates respond to shocks to lending supply and borrowing demand while also ensuring that markets clear. For this reason, the DeFi lending market equilibrium is always inefficient relative to a market-clearing competitive equilibrium.

We then study the optimal design of the PLF interest rate function. In the baseline case without withdrawal demand uncertainty, we show that it is possible to generate DeFi equilibrium interest rates and efficiency arbitrarily close to that of a competitive equilibrium. In particular, although the PLF's interest rates only adjust when the utilization rate is below unity, setting a PLF interest rate function with a steep slope ensures that the PLF interest rates are sufficiently reactive to changes in the utilization rate, which guarantees

that equilibrium utilization remains close to unity.³

In deriving the approximately optimal design, we first show that the optimal interest rate function should set sufficiently low interest rates for low levels of utilization. When this is done, then those sufficiently low rates will guarantee an immediate increase in borrowing demand and a decrease in lending supply when utilization is low, forcing equilibrium utilization up and eliminating excess demand. Importantly, as a second step, the optimal design must also increase the slope of the interest rate function as utilization approaches unity; otherwise, these low interest rates would produce a utilization of 100%. As mentioned above, when utilization reaches 100%, interest rates no longer become responsive to market conditions, which leads to excess supply. By setting a large slope, it is possible to ensure that the PLF interest rates are sufficiently high when utilization is near unity. This feature guarantees that equilibrium utilization is always less than 100% as borrowers no longer wish to borrow at such high rates, preventing excess supply. Hence, the optimal interest rate function design sets a high slope to mitigate excess supply and low rates for low levels of utilization to mitigate excess demand. In fact, without uncertainty about withdrawal demand, setting the steepest possible slope is optimal for the platform as it allows for the minimization of efficiency loss without any cost. This feature also explains why any such design is only approximately optimal, as the PLF interest rate function must set a finite slope. While this result represents a useful benchmark, underscoring the economics of DeFi lending platforms, it is, however, unlikely to hold in the more realistic scenario when market participants' withdrawals introduce noise in the utilization rate.

To capture this additional uncertainty, we assume that, with some probability, a fraction of lenders will need to withdraw the funds that they lent to the platform shortly after the borrowing and lending decisions are made. This assumption allows us to demonstrate an

³Note that this design can be adjusted to ensure instead that equilibrium utilization remains sufficiently close to any target less than unity. We discuss such utilization targets and their optimality in the presence of withdrawal shocks in Section 5.

important trade-off in the interest rate design in a tractable manner. Namely, when lenders are shocked and withdraw their funds, the utilization rate increases and pushes interest rates up along the PLF’s positive slope within the same period. The steeper the slope of the PLF function, the larger those withdrawal demand shocks translate into increased interest rates to be paid by the borrowers and, thus, the larger volatility of PLF interest rates. Because borrowers are averse to interest rate risk, a steeper slope is not necessarily beneficial anymore. In contrast, the platform must consider the trade-off between efficiency and interest rate volatility when determining its optimal PLF interest rate function design. We characterize the optimal slope and intercept of linear interest rate functions and show the existence of an optimal interest rate function whose slope balances the aforementioned efficiency-volatility trade-off.

Literature Our paper contributes to the literature that studies the economics of blockchain technology. Much of the early work in that literature examines economic security (see, e.g., [Biais et al. 2019](#), [Easley et al. 2019](#), and [Saleh 2021](#)). In contrast, we examine a secure blockchain with smart contract functionality (e.g., Ethereum) and study welfare implications arising from a prominent application, DeFi lending, facilitated by a PLF. The economics literature that studies blockchains with smart contract functionality is recent and growing (see e.g., [Capponi et al. 2023](#) and [John et al. 2023](#)) - topics of study within this literature include the economics of digital tokens (see, e.g., [Cong et al. 2021](#) and [Mayer 2022](#)), stablecoins (see, e.g., [d’Avernas et al. 2022](#) and [Li and Mayer 2022](#)), and decentralized exchanges (see, e.g., [Capponi and Jia 2021](#), [Hasbrouck et al. 2023](#), [Lehar and Parlour 2024](#), and [Park 2021](#)). We add to the smart contracts literature by providing a formal analysis of a PLF with a particular focus on the PLF’s interest rate-setting mechanism. Other notable papers that also examine PLFs include [Aramonte et al. \(2022\)](#), [Carré and Gabriel \(2022\)](#), [Chaudhary et al. \(2023\)](#), [Chiu et al. \(2022\)](#), [Lehar and Parlour \(2022\)](#), [Mueller \(2022\)](#), and [Park and](#)

Stinner (2023). Aramonte et al. (2022) highlight that PLFs serve largely as a vehicle for leveraged cryptoasset trading. Contemporaneously, Carré and Gabriel (2022) study the design of the interest rate setting mechanism focusing on the role that the security of the underlying blockchain plays when determining the interest rate function. Chiu et al. (2022) theoretically model a PLF while focusing on asymmetric information between borrowers and lenders. Chaudhary et al. (2023), Lehar and Parlour (2022), and Mueller (2022) provide empirical insights regarding PLFs. Park and Stinner (2023) demonstrate how certain incentive programs affect borrowing and lending decisions at DeFi lending platforms. Motivating the withdrawal uncertainty presented in this paper, Park and Stinner (2023) demonstrate how these incentive programs create “yield chasing” strategies that lead to large, and difficult to predict, fluctuations in borrowing and lending rates when these programs are phased in and out.

Our work differs from prior DeFi lending papers in that we focus on the interest rate-setting mechanism and the associated welfare implications. Aramonte et al. (2022) point out that the mechanics of a PLF can also be applied to non-cryptoassets through the tokenization of real assets and that such tokenization could have important welfare implications for SMEs. Our analysis formalizes that assertion, clarifying the welfare implications of PLFs if integrated with traditional finance (see also John et al. 2023).

2 Model

We examine a discrete time infinite horizon model in which time is indexed by $t \in \mathbb{N}$. At the beginning of each period, a random measure of lenders and borrowers arrive at a DeFi lending market, which consists of a single PLF. Lenders possess a unit of capital and heterogeneous investment opportunities so that each lender may either lend to the PLF or invest in her alternative investment opportunity instead. Borrowers possess no capital but

have heterogeneous investment opportunities which they finance with borrowed funds if they find it optimal to do so.⁴ We assume that the PLF sets both the borrower's and the lender's interest rates as per practice. Lenders and borrowers decide whether to interact with the PLF (by lending or borrowing, respectively), knowing in advance how those interest rates are determined through the PLF. At the end of each period, each borrower who borrows from the PLF repays her loan with interest, and the PLF channels those payments to remunerate the lenders who lent to the PLF. The next period then commences with a new random measure of borrowers and lenders arriving at the DeFi lending market.

2.1 Market Participants

Arrival Process At the beginning of each period, a measure λ_t of lenders and a measure μ_t of borrowers arrive where both $\{\lambda_t\}_{t=1}^\infty$ and $\{\mu_t\}_{t=1}^\infty$ constitute strictly positive sequences with i.i.d elements and finite first moments. As we shall see, λ_t and μ_t affect equilibrium solutions only through their ratio, μ_t/λ_t , so that it is convenient to define an additional random sequence, $\{\theta_t\}_{t=1}^\infty$, as:

$$\theta_t := \frac{\mu_t}{\lambda_t}. \quad (2.1)$$

Note that θ_t captures the credit market tightness from the borrower's perspective with high values of θ_t corresponding to high market tightness — relatively more borrowers as compared to lenders (i.e., high μ_t or low λ_t). Consequently, we hereafter refer to θ_t as the time t *market tightness*. Moreover, as a simplifying regularity condition, we require that θ_t possesses a strictly increasing and continuously differentiable cumulative distribution function, G , which is supported on $(0, +\infty)$. We uniquely identify each lender by the ordered pair $(j, t) \in [0, \lambda_t] \times \mathbb{N}$ and each borrower by the order pair $(i, t) \in [0, \mu_t] \times \mathbb{N}$ where t denotes the period

⁴Note that we abstract from collateral given that the focus of this paper is on the PLF interest rate mechanism, which is orthogonal to the collateral mechanism. All of our results would hold if we required borrowers to post collateral to secure their loans.

in which the market participant arrives, j denotes the index of each lender among lenders, and i denotes the index of the borrower among borrowers.

Lenders For each $j \in [0, \lambda_t]$ we assume that Lender (j, t) possesses a unit of capital and has access to an investment opportunity with net expected return $r_{l,(j,t)} \sim F_l[0, +\infty)$ where $F_l \in \mathcal{C}^1[0, +\infty)$ denotes a continuously differentiable and strictly increasing cumulative distribution function with a finite first moment. We use the convention that $F_l(x) := 0$ for all $x < 0$. For each period t , Lender (j, t) may invest her capital in the PLF or invest it in her alternative investment opportunity.

Borrowers For each $i \in [0, \mu_t]$ we assume that Borrower (i, t) possesses no capital and has access to an investment opportunity with net expected return $r_{b,(i,t)} \sim F_b[0, +\infty)$ where $F_b \in \mathcal{C}^1[0, +\infty)$ denotes a continuously differentiable and strictly increasing cumulative distribution function with a finite first moment. Again, we use the convention that $F_b(x) := 0$ for all $x < 0$. We assume that Borrower (i, t) may invest up to one unit of capital in her investment opportunity.

2.2 Withdrawal Uncertainty

Withdrawal Shock At the beginning of each period t , lenders and borrowers make their decisions, anticipating the possibility of a subsequent shock to lending supply. Immediately after borrowing and lending decisions have been made, lenders receive a withdrawal shock with probability q , requiring a fraction $\eta \in [0, 1)$ of them to withdraw their capital from the platform for alternative uses. After a withdrawal shock is realized, the borrowers and lenders who do not receive a withdrawal shock are unable to alter their borrowing or lending decisions until the next period $t + 1$. This assumption simplifies the analysis while capturing an important feature of DeFi lending: it is not always possible (nor optimal) for lenders and

borrowers to instantaneously adjust their positions following a shock. Finally, given that we assume that the withdrawal shock is realized immediately after lending and borrowing decisions are made, the lending and borrowing rates paid in period t correspond to those that arise after the withdrawal shock is drawn.

Discussion of Assumption Withdrawal shocks allow us to capture two important features of PLFs: (i) lenders may need to withdraw their loanable funds early but can only do so if there are sufficient funds available (i.e., the utilization rate is sufficiently below one) and (ii) utilization rates can be volatile due to exogenous noise in supply and demand.⁵ While we could capture those effects through a more generic shock to borrowing and lending, assuming a single shock to lenders is without loss of generality.⁶ In Sections 3 and 4, we first characterize the properties of a DeFi lending equilibrium in the baseline model without withdrawal shocks, which we then reintroduce in Section 5.

2.3 Lending Supply and Borrowing Demand

Lending Supply Note that if the utilization rate of the PLF is larger than $1 - \eta$ at time t , then some lenders will be unable to withdraw their capital upon a shock realization. In this case, we assume that any lender that cannot withdraw their full unit of capital faces a cost $\kappa > 0$. Thus, denoting by $\mathbb{E}[l_t] \geq 0$, the expected PLF lending rate in period t , then for each $j \in [0, \lambda_t]$, Lender (j, t) lends to the PLF at time t whenever the return from their outside option is lower than the expected return from lending to the platform, adjusting for

⁵Note that all PLF loans are perpetual and all loanable funds can be redeemed on demand provided that there are sufficient funds available at the time of redemption.

⁶Note also that the above assumption that borrowers and (non-shocked) lenders do not adjust their borrowing/lending decisions after a shock realizes within a period is also without loss of generality. In particular, we demonstrate below that withdrawal shocks effectively represent a change from the state θ_t to the state $\theta_t/(1 - \eta)$. Therefore, allowing all agents to optimally adjust their borrowing and lending decisions after such shocks would still produce a positive variance in PLF interest rates in each period, which is the key feature that the withdrawal shocks are intended to generate.

the expected cost of not being able to withdraw:

$$r_{l,(j,t)} \leq \mathbb{E}[l_t] - \hat{\kappa} \cdot \mathbb{1}(U_t > 1 - \eta). \quad (2.2)$$

where we simplify our notation by denoting $\hat{\kappa} := q \cdot \eta \cdot \kappa$ as the expected cost of not being able to withdraw funds when the withdrawal shock is realized.⁷ Given (2.2), and the fact that $r_{l,(j,t)} \sim F_l$, the beginning of period t supply for the PLF, $\mathcal{S}_t$, must satisfy the following condition:

$$\mathcal{S}_t = \lambda_t \cdot F_l(\mathbb{E}[l_t] - \hat{\kappa} \cdot \mathbb{1}(U_t > 1 - \eta)). \quad (2.3)$$

Finally, note the expected utility of Lender (j, t) , $\mathcal{U}_{(j,t)}^l$, is given as

$$\mathcal{U}_{l,(j,t)} := \max\{\mathbb{E}[l_t] - \hat{\kappa} \cdot \mathbb{1}(U_t > 1 - \eta), r_{l,(j,t)}\}.$$

Borrowing Demand We assume that borrowers are risk averse with mean-variance utility over the PLF interest rate volatility. Thus, we write Borrower (i, t) 's utility per unit of capital borrowed (and invested in their investment opportunity with return $r_{b,(i,t)}$) as

$$r_{b,(i,t)} - \mathbb{E}[b_t] - \zeta \cdot \text{VAR}(b_t),$$

where $\mathbb{E}[b_t]$ is the expected borrowing rate that the borrower pays in period t (given the anticipated withdrawal shocks), $\text{VAR}(b_t)$ is the variance of the borrower rate that arises due to the realized withdrawal shocks, and ζ is a parameter capturing the borrower's risk aversion.⁸ These borrower preferences imply that for each $i \in [0, \mu_t]$, Borrower (i, t) optimally

⁷Note that in deriving (2.2), we implicitly assume that each shocked lender receives the same return from activities after redemption as the return that they would receive in the state whereby the shock occurs, but that lender was not shocked (and thus continues lending to the platform). This assumption greatly simplifies the analysis and is without loss when compared to specifying an explicit payoff for shocked lenders.

⁸Note that our results will hold when assuming that lenders are equally risk-averse.

borrowers one unit of capital from the PLF if and only if $r_{b,(i,t)} \geq \mathbb{E}[b_t] + \zeta \cdot \text{VAR}(b_t)$. As a consequence, the beginning of period t demand for the PLF, $\mathcal{D}_t$, must satisfy the following condition:

$$\mathcal{D}_t = \mu_t \cdot (1 - F_b(\mathbb{E}[b_t] + \zeta \cdot \text{VAR}(b_t))). \quad (2.4)$$

As a practical matter, the PLF cannot lend out funds beyond what has been deposited by lenders. In particular, whenever borrowing demand exceeds lending supply (i.e., $\mathcal{D}_t > \mathcal{S}_t$), the PLF cannot fulfill all borrowing demand. In turn, we assume that the PLF lends out its funds pro-rata whenever borrowing demand exceeds lending supply, implying that each borrower seeking a unit of capital receives the fraction $\mathcal{S}_t/\mathcal{D}_t$ in that case. Then, the utility of Borrower (i, t) , $\mathcal{U}_{b,(i,t)}$, is given as:

$$\mathcal{U}_{b,(i,t)} := \begin{cases} \max\{r_{b,(i,t)} - \mathbb{E}[b_t] - \zeta \cdot \text{VAR}(b_t), 0\} & \text{if } \mathcal{D}_t \leq \mathcal{S}_t \\ \max\{\frac{\mathcal{S}_t}{\mathcal{D}_t} \cdot (r_{b,(i,t)} - \mathbb{E}[b_t] - \zeta \cdot \text{VAR}(b_t)), 0\} & \text{if } \mathcal{D}_t > \mathcal{S}_t. \end{cases} \quad (2.5)$$

2.4 Protocol for Loanable Funds

PLF Interest Rate Functions A PLF is defined by an exogenous interest rate function, $\rho : [0, 1] \mapsto \mathbb{R}$, which is continuously differentiable, non-decreasing, and satisfies $\rho(U) > 0$ for some $U \in [0, 1]$. As discussed above, we define the utilization rate, U_t , as the proportion of funds lent to the PLF that have been borrowed; more explicitly, the utilization rate, U_t , at any point in time when borrower demand is $\mathcal{D}_t$ and lending supply is $\mathcal{S}_t$ is given by

$$U_t = \frac{\text{Total Borrowing}}{\text{Total Lending}} = \frac{\min\{\mathcal{D}_t, \mathcal{S}_t\}}{\mathcal{S}_t} = \min\{\frac{\mathcal{D}_t}{\mathcal{S}_t}, 1\}, \quad (2.6)$$

where the minimum operator in the numerator of Equation (2.6) arises because, as previously discussed, the PLF cannot lend out funds in excess of those received from lenders. For

convenience, we utilize the convention that when $\mathcal{S}_t = 0$ and $\mathcal{D}_t > 0$ then $U_t = 1$. Fixing the exogenous interest rate function, ρ , whenever the utilization rate is U_t , the PLF sets the borrower's interest rate, b_t , as

$$b_t = \rho(U_t). \quad (2.7)$$

PLF Lending Rate Turning to the procedure for setting the PLF lender rate, the PLF passes borrower payments directly onto lenders, thereby implying that if the borrowing rate is b_t , the PLF sets the lender interest rate l_t to solve:

$$\underbrace{\mathcal{S}_t \cdot l_t}_{\text{payments to lenders}} = \underbrace{\min\{\mathcal{S}_t, \mathcal{D}_t\} \cdot b_t}_{\text{payments from borrowers}}. \quad (2.8)$$

In turn, dividing both sides of Equation (2.8) by total lending, $\mathcal{S}_t$, and applying the definition of the utilization rate in Equation (2.6) yields the PLF's mechanical rule for setting the lender interest rate:

$$l_t = U_t \cdot \rho(U_t). \quad (2.9)$$

Utilization Post-Withdrawal Recall that U_t denotes the utilization rate at the start of period t before the withdrawal shock is drawn and denote by U_t^+ the utilization rate that arises in period t after the withdrawal shock is realized. Then, using the above expressions, the expected borrower rate in period t , when the utilization rate at the start of the period is U_t , is given as by

$$\mathbb{E}[b_t] = (1 - q) \cdot \rho(U_t) + q \cdot \rho(U_t^+). \quad (2.10)$$

Similarly, the expected period t lender rate is given by

$$\mathbb{E}[l_t] = (1 - q) \cdot U_t \cdot \rho(U_t) + q \cdot U_t^+ \cdot \rho(U_t^+). \quad (2.11)$$

Finally, note that the assumption that only shocked lenders adjust to the shock after the beginning of time t implies that for any demand and supply for loanable funds at the start of period t , $\mathcal{D}_t$ and $\mathcal{S}_t$, the demand after a withdrawal shock is realized remains the same, equal to $\mathcal{D}_t$, while the supply after a withdrawal shock is realized becomes $(1 - \eta) \cdot \mathcal{S}_t$. This implies that when the utilization rate is U_t at the start of the period, and the withdrawal shock is subsequently realized, then the utilization rate becomes $U_t^+ = \min\{U_t/(1 - \eta), 1\}$.

2.5 Equilibrium

Equilibrium Definition A DeFi lending equilibrium in period t consists of the expected borrower and lender rates $\mathbb{E}[b_t], \mathbb{E}[l_t]$, and a utilization rate U_t . To determine the equilibrium utilization rate at the start of period t , U_t^* , we apply the borrowing rate rule in Equation (2.7) and the lending rate rule in Equation (2.9) to the definition of the utilization rate in Equation (2.6) and use the fact that $U_t^+ = \min\{U_t/(1 - \eta), 1\}$. Furthermore, using the notation that $\mathbb{E}[b_t] = \mathbb{E}[\rho(U_t)]$ and $\mathbb{E}[l_t] = \mathbb{E}[U_t \rho(U_t)]$ then using (2.10), we denote by

$$VAR(b_t) = (1 - q) \cdot (\mathbb{E}_t[\rho(U_t)] - \rho(U_t))^2 + q \cdot (\mathbb{E}_t[\rho(U_t)] - \rho(U_t^+))^2, \quad (2.12)$$

the variance of interest rates at the start of period t given that borrowers at the start of the period expect to pay the interest rate $\mathbb{E}[\rho(U_t)]$ but ultimately pay $\rho(U_t)$ when there is no withdrawal shock (with probability $(1 - q)$) and pay $\rho(U_t^+)$ when there is a withdrawal shock (with probability q). Further note that for any $U \in [0, 1]$ the following equations hold:

$$\mathbb{E}[\rho(U_t)] := (1 - q) \cdot \rho(U_t) + q \cdot \rho(\min\{\frac{U_t}{1 - \eta}, 1\}), \quad (2.13)$$

$$\mathbb{E}[U_t \rho(U_t)] := (1 - q) \cdot U_t \cdot \rho(U_t) + q \cdot \min\{\frac{U_t}{1 - \eta}, 1\} \cdot \rho(\min\{\frac{U_t}{1 - \eta}, 1\}). \quad (2.14)$$

The equilibrium utilization rate at the start of period t is determined as the solution to the following fixed point problem:

$$U_t^* = \min\left\{ \frac{\mathcal{D}_t(\mathbb{E}_t[\rho(U_t)])}{\mathcal{S}_t(\mathbb{E}_t[U_t \cdot \rho(U_t)])}, 1 \right\}, \quad (2.15)$$

where we explicitly note the dependence of $\mathcal{S}_t$ and $\mathcal{D}_t$ on the interest rate function ρ as given by (2.3) and (2.4) when using the fact that $\mathbb{E}[b_t] = \mathbb{E}[\rho(U_t)]$ and $\mathbb{E}[l_t] = \mathbb{E}[U_t \cdot \rho(U_t)]$. That is, a DeFi Lending equilibrium is a fixed point in which utilization is consistent with the market participants' optimal lending and borrowing decisions when taking into account that the PLF sets interest rates as a function of utilization.

3 Baseline Model Analysis

For the first part of our analysis, we study a baseline case in which the fraction of lenders that receive withdrawal shocks in each period is equal to zero (i.e., $\eta = 0$). We postpone the analysis of the effect of this additional factor to Section 5. Until then, our baseline case implies that there is a single utilization rate per period which is determined by the following equation

$$U_t = \min\left\{ \theta \cdot \frac{1 - F_b(\rho(U_t))}{F_l(U_t \cdot \rho(U_t))}, 1 \right\}. \quad (3.1)$$

In addition, note that in this case $\mathbb{E}[b_t] = b_t = \rho(U_t)$ and $\mathbb{E}[l_t] = l_t = U_t \cdot \rho(U_t)$ implying that there is no noise in lending and borrowing rates and thus in what follows we drop the expectation operator on b_t and l_t .

3.1 Existence and Uniqueness of a DeFi Lending Equilibrium

Our first proposition demonstrates that the solution to (3.1) always exists and is unique.

Proposition 3.1. DeFi Lending Equilibrium

A unique DeFi lending equilibrium always exists. Within this equilibrium, in each period t , the borrower interest rate b_t , the lender interest rate l_t , and the utilization rate U_t depend only on the market tightness $\theta_t = \mu_t/\lambda_t$ with the equilibrium solutions being given by:

$$U_t^* = U^*(\theta_t), \quad b_t^* = \rho(U^*(\theta_t)), \quad l_t^* = U^*(\theta_t) \cdot \rho(U^*(\theta_t)), \quad (3.2)$$

where $U^* : [0, +\infty) \mapsto [0, 1]$ is the point-wise unique function satisfying:

$$U^*(\theta) = \min \left\{ \theta \cdot \frac{1 - F_b(\rho(U^*(\theta)))}{F_l(U^*(\theta) \cdot \rho(U^*(\theta)))}, 1 \right\}. \quad (3.3)$$

Proposition 3.3 demonstrates that PLFs generate a stable and unambiguous economic outcome. This result is obtained because a PLF adjusts interest rates according to market tightness (i.e., b_t^* and l_t^* vary with μ_t and λ_t through θ_t). In particular, since supply and demand curves are not directly observable in practice, a PLF employs the utilization rate as a proxy for market tightness and sets interest rates as an increasing function of the utilization rate (see Equation 2.7 and 2.9). In turn, increased borrowing demand (i.e., a larger μ_t) increases PLF interest rates because more demand implies a higher utilization rate, which then mechanically increases PLF interest rates. Similarly, an increase in lending supply (i.e., increases in λ_t) implies a decrease in PLF interest rates through a reduction in the utilization rate, which then mechanically decreases PLF interest rates. We formalize this reasoning with the following result.

Proposition 3.2. Lending, Borrowing, and Utilization Rates Increase in Market Tightness

If $\theta_t < \theta_{t'}$, then the following results hold:

$$U_t^* \leq U_{t'}^*, \quad b_t^* \leq b_{t'}^*, \quad l_t^* \leq l_{t'}^* \quad (3.4)$$

Additionally, if ρ is strictly increasing and $U_t^* < 1$, then the above inequalities are all strict.

3.2 Welfare

Having established the existence and uniqueness of a DeFi lending equilibrium, we now examine how welfare in this DeFi equilibrium differs from a competitive equilibrium. We define welfare at time t as the sum of the expected utility of the borrowers and lenders at the start of period t . As a consequence, the expected period t welfare in a DeFi lending equilibrium, $\mathcal{W}_t^{DeFi}$, is given by

$$\mathcal{W}_t^{DeFi} := \underbrace{\int_0^{\mu_t} \mathcal{U}_{b,(i,t)}^* di}_{\text{Borrower Welfare}} + \underbrace{\int_0^{\lambda_t} \mathcal{U}_{l,(j,t)}^* dj}_{\text{Lender Welfare}}, \quad (3.5)$$

where $\mathcal{U}_{b,(i,t)}^*$ and $\mathcal{U}_{l,(j,t)}^*$ are the DeFi equilibrium utilities of Borrower (i, t) and Lender (j, t) respectively when $\mathbb{E}[b_t^*] = b_t^* = \rho(U_t^*)$ and $\mathbb{E}[l_t^*] = l_t^* = U_t^* \cdot \rho(U_t^*)$. In contrast, the period t welfare in a competitive lending equilibrium, $\mathcal{W}_t^{CE}$, is given by the sum of utility of the borrowers and lenders in the case in which the borrowing and lending rates are both equal to the market-clearing interest rate, r_t^C . Note that r_t^C is defined as the rate that equates lending supply and borrowing demand: $\mathcal{D}_t(r_t^C) = \mathcal{S}_t(r_t^C)$. Therefore,

$$\mathcal{W}_t^{CE} := \underbrace{\int_0^{\mu_t} \mathcal{U}_{b,(i,t)}^C di}_{\text{Borrower Welfare}} + \underbrace{\int_0^{\lambda_t} \mathcal{U}_{l,(j,t)}^C dj}_{\text{Lender Welfare}}, \quad (3.6)$$

where $\mathcal{U}_{b,(i,t)}^C$ and $\mathcal{U}_{l,(j,t)}^C$ respectively denote the borrower and lender utility when rates are set to r_t^C (i.e., $b_t = l_t = r_t^C$) in a competitive lending market.

As a preliminary result, we establish that the competitive equilibrium interest rate, r_t^C , always exists, is unique, and is increasing in the credit market tightness θ_t :

Proposition 3.3. The Competitive Equilibrium Interest Rate is Unique

A unique competitive lending equilibrium always exists. Within this equilibrium, in each period t , the borrower interest rate b_t^C , the lender interest rate l_t^C , and the utilization rate U_t^C only depend on the market tightness $\theta_t = \mu_t/\lambda_t$ with the equilibrium solutions being given explicitly as follows:

$$U_t^C = 1 \quad b_t^C = l_t^C = r_t^C = r^C(\theta_t),$$

with $r^C : [0, +\infty) \mapsto [0, +\infty)$ the unique function satisfying:

$$F_l(r^C(\theta_t)) = \theta_t \cdot (1 - F_b(r^C(\theta_t))), \quad (3.7)$$

which is guaranteed to exist. Furthermore, r^C is one-to-one, onto, and strictly increasing in the market tightness θ_t .

3.3 The PLF identification Problem

We have shown in Proposition 3.3 that for any level of market tightness θ_t there exists a unique market clearing rate $r^C(\theta_t)$. Our next result illustrates the PLF identification problem, which precludes the PLF from setting the competitive equilibrium interest rate for each possible realization of market tightness θ_t . In order to highlight the generality of this identification problem, we present this result for a general class of PLF interest rate functions which map the realized supply, $\mathcal{S}_t$, and demand, $\mathcal{D}_t$, to an interest rate $\rho(\mathcal{S}_t, \mathcal{D}_t)$ for each period.

Proposition 3.4. PLF Identification Problem

Consider a general PLF interest rate function $\rho : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ that sets an interest rate, $\rho(\mathcal{S}_t, \mathcal{D}_t)$, as a function of the realized supply $\mathcal{S}_t$ of loanable funds and borrowing demand $\mathcal{D}_t$ in each period. Further assume that μ_t and λ_t are continuously distributed. Then, for any

such function ρ , it must be the case that $Pr(\rho(\mathcal{S}_t, \mathcal{D}_t) = r_t^C) = 0$.

This result is the consequence of an inherent constraint facing DeFi lending platforms that rely on a PLF and is reminiscent of the classical identification problem in the presence of simultaneous shocks in econometrics. Given that μ_t and λ_t are both simultaneously random, it is impossible for the realized supply and demand to identify the market clearing rate.

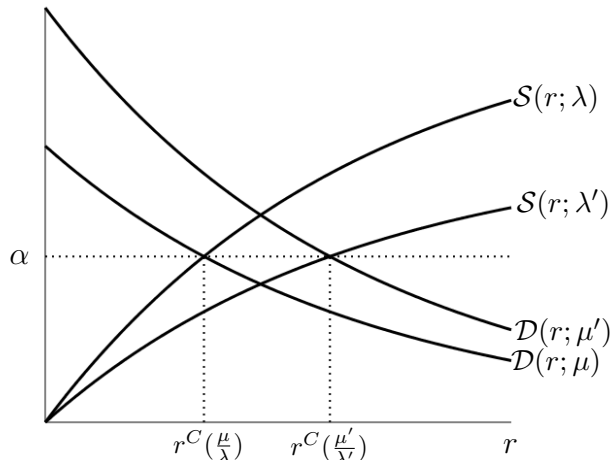


Figure 1: An illustration of the PLF identification problem when $F_l(x) = F_b(x) = 1 - e^{-15x}$ for two different pairs of market conditions $(\mu, \lambda) = (.4, .6)$ and $(\mu', \lambda') = (.6, .4)$.

We illustrate this identification problem in Figure 1 in which we plot the supply and demand as a function of the interest rate r under two different pairs of market conditions $(\mu, \lambda) = (.4, .6)$ and $(\mu', \lambda') = (.6, .4)$. As illustrated, the market clearing rates, $r^C(\mu/\lambda)$ and $r^C(\mu'/\lambda')$. Both market conditions (μ, λ) and (μ', λ') are such that supply equals demand at α . An efficient PLF would like to invert this relation and systematically specify a unique interest rate consistent with the lending market clearing at α . However, as can be seen from Figure 1, it is not possible to always select the right market clearing rate when only observing that realized supply equals realized demand at α without knowing whether the market condition is (μ, λ) or (μ', λ') as $r^C(\mu/\lambda) \neq r^C(\mu'/\lambda')$. Although we illustrate this principle for only two states (μ, λ) and (μ', λ') , a similar argument applies for infinitely

many pairs, each of which corresponds to a specific market clearing rate. Hence, when arbitrarily specifying an interest rate function, the probability of setting the correct market clearing rate as a function of equilibrium quantities is zero.

We note here that by deriving Proposition 3.4 under a general PLF interest rate function, our intention was to demonstrate that the identification problem is not limited to taking the utilization ratio as an input but rather applies to any unique observables that would be available on the blockchain, making it a fundamental feature of DeFi lending platform designs. In what follows, we revert back to studying the class of PLF interest rate functions presented in Section 2.4.

3.4 Inefficiency of DeFi Lending Platforms

As demonstrated in the previous section, the PLF identification problem generates an informational disadvantage for PLFs when compared to a competitive market. In the following proposition, we show that this disadvantage results in reduced welfare in a DeFi equilibrium when compared to a competitive equilibrium.

Proposition 3.5. *Inefficiency of DeFi Lending Equilibrium*

For any level of the market tightness θ_t , the DeFi lending equilibrium generates lower welfare than the competitive lending equilibrium so that $\mathcal{W}_t^{DeFi} \leq \mathcal{W}_t^{CE}$ for all t .

The DeFi lending equilibrium is sub-optimal because welfare is determined entirely by interest rates (see Equations 3.5 and 3.6), and the PLF is not able to perfectly match the competitive market equilibrium interest rate for all realizations of the market tightness. Importantly, we prove in our next result that the PLF interest rate function can match the competitive equilibrium lending rates and welfare for only a single realization of market tightness θ_t .

Proposition 3.6. DeFi Equilibrium Matches the CE at a Unique θ_t

There exists a unique realization of the market tightness θ_t such that the DeFi equilibrium matches the competitive lending equilibrium in period t . More formally, there exists a unique $\tilde{\theta}$ such that:

$$\mathcal{W}_t^{DeFi} = \mathcal{W}_t^{CE} \text{ if and only if } \theta_t = \tilde{\theta}. \quad (3.8)$$

For this value of $\tilde{\theta}$, PLF interest rates and the competitive equilibrium interest rate all equal $\rho(1)$:

$$r_t^C = b_t^* = l_t^* = \rho(1) \iff \theta_t = \tilde{\theta}. \quad (3.9)$$

The aforementioned unique value of $\tilde{\theta}$ is given explicitly as follows:

$$\tilde{\theta} = (r^C)^{-1}(\rho(1)), \quad (3.10)$$

where $(r^C)^{-1}$ refers to the inverse function of r^C .

Note that Proposition 3.6 holds for *any* interest rate function ρ . Therefore, independently from the choice of the interest rate function ρ , there always exists a unique level of market tightness, $\tilde{\theta}$, whereby the interest rate function ρ generates the highest possible level of welfare and thus ρ generates lower welfare for all other θ . To understand Proposition 3.6, note that in order for both the PLF borrower and lender interest rates, b_t^* and l_t^* , to be equal to the competitive equilibrium interest rate, r_t^C , it must be that PLF borrower and lender interest rates are identical. However, per Equations (2.7) and (2.9), the PLF borrower and lender interest rates are identical only when the PLF achieves full utilization (i.e., when $U_t^* = 1$). Since the PLF sets interest rates as a mechanical function of the utilization rate, full utilization sets both the PLF borrower and lender interest rates at $\rho(U_t^*) = \rho(1)$. Consequently, the DeFi lending equilibrium achieves the competitive lending equilibrium welfare

only if the competitive equilibrium interest rate, r_t^C , equals $\rho(1)$. Finally, since the competitive equilibrium interest rate, r_t^C , is an increasing function of the market tightness θ_t (see Proposition 3.3), the competitive equilibrium interest rate, r_t^C , and the PLF interest rates correspond exclusively for the unique realization of the market tightness $\theta_t = \tilde{\theta}$ satisfying $r^C(\tilde{\theta}) = \rho(1)$.

To better understand the inefficiency within DeFi lending markets, the following additional result clarifies how PLF interest rates deviate from the competitive equilibrium interest rates when the market tightness is such that the PLF does not achieve the first-best efficiency (i.e., when $\theta_t \neq \tilde{\theta}$ with $\tilde{\theta}$ the value given by (3.10) in Proposition 3.6):

Proposition 3.7. *DeFi Interest Rates Deviate From the CE*

When the market tightness θ_t entails either an excess of borrowers or a shortage of lenders relative to $\tilde{\theta}$ (i.e., when $\theta_t > \tilde{\theta}$), the competitive equilibrium interest rate, r_t^C , strictly exceeds both the PLF borrower interest rate, b_t^ , and the PLF lender interest rate, l_t^* :*

$$\theta_t > \tilde{\theta} \implies r_t^C > b_t^* \text{ and } r_t^C > l_t^*. \quad (3.11)$$

When the market tightness θ_t entails either a shortage of borrowers or an excess of lenders relative to $\tilde{\theta}$ (i.e., when $\theta_t < \tilde{\theta}$), then the PLF borrower interest rate, b_t^ , strictly exceeds the competitive equilibrium interest rate, r_t^C :*

$$\theta_t < \tilde{\theta} \implies b_t^* > r_t^C, \quad (3.12)$$

whereas the relationship between the competitive equilibrium interest rate, r_t^C , and the PLF lender interest rate, l_t^ , is ambiguous.*

The first part of Proposition 3.7 establishes that both the PLF borrower interest rate, b_t^* , and PLF lender interest rate, l_t^* , are strictly lower than the competitive equilibrium

interest rate, r_t^C , when borrowing demand is high or lending supply is low relative to $\tilde{\theta}$ (i.e., when $\theta_t > \tilde{\theta}$). In contrast, the second part of Proposition 3.7 demonstrates that the PLF borrower interest rate, b_t^* , strictly exceeds the competitive equilibrium interest rate, r_t^C , when borrowing demand is low or lending supply is high relative to $\tilde{\theta}$ (i.e., when $\theta_t < \tilde{\theta}$). In the latter case, the relationship between the competitive equilibrium interest rate, r_t^C , and the PLF lender interest rate, l_t^* , is ambiguous.

4 Optimal PLF Design in the Baseline Model

In this section, we study the optimal design of the PLF interest rate function, ρ , in the baseline model without withdrawal shocks. We show that, although it is not possible for the DeFi equilibrium to achieve first-best, it is possible to design the interest rate function to ensure that the DeFi equilibrium is arbitrarily close to it. In this sense, the optimally designed PLF is *approximately efficient*. As we focus on optimal PLF design, which is time-invariant, we drop the time sub-scripts in what follows to economize on notation.

4.1 An Approximately Efficient PLF

We first show that it is possible to design the interest rate function ρ such that the equilibrium utilization rate, $U^*(\theta)$, is arbitrarily close (but not equal) to unity for all realizations of market tightness θ .

Proposition 4.1. Targeting Utilization Near Unity

Suppose that $\theta_t \sim G[\underline{\theta}, +\infty)$ where $\underline{\theta} > 0$. Then, for any sufficiently small $\epsilon > 0$ there exists an interest rate function ρ such that $U^(\theta) \in [1 - \epsilon, 1)$ for all $\theta \in [\underline{\theta}, +\infty)$.*

We derive the result of Proposition 4.1 by constructing the interest rate function ρ that guarantees $U^*(\theta) \in [1 - \epsilon, 1)$ for all $\theta \in [\underline{\theta}, +\infty)$. The design of this interest rate function

requires setting $\rho(U)$ to be sufficiently small for all $U < 1 - \epsilon$ to induce higher levels of borrowing and lower levels of lending through lower rates. Therefore, any utilization rate $U < 1 - \epsilon$ does not satisfy our equilibrium condition for any state, and thus $U^*(\theta) \geq 1 - \epsilon$ for all $\theta \in [\underline{\theta}, +\infty)$. This construction ensures that excess demand decreases as ϵ decreases.

Importantly, this simple step of choosing lower interest rates for all U is not sufficient to improve welfare. In particular, a subtle feature of Proposition 4.1 is that we require that ρ is set to ensure that utilization is never equal to unity. This feature is crucial because it addresses an issue arising when reducing $\rho(U)$ for each $U \in [0, 1]$ due to the potential of reaching the upper limit, $U^*(\theta) = 1$, which occurs when excess demand reaches zero. Upon reaching this limit, a new inefficiency arises due to excess supply as $U^*(\theta) = 1$ implies $U^*(\theta') = 1$ for all $\theta' \geq \theta$ making PLF interest rates inelastic at $\rho(1)$. To resolve this issue, the approximately optimal interest rate function must also increase its slope as it lowers the interest rates for all $U \in [0, 1]$ so that for any θ , there exists a $U \in [1 - \epsilon, 1)$ such that $\rho(U) = r^C(\theta)$. This property guarantees that, as utilization approaches unity, the borrower interest rate increases by so much that borrowers no longer find it optimal to borrow at those rates, bounding the utilization rate away from 1. Therefore, $U = 1$ cannot be an equilibrium when setting $\rho(1)$ to be sufficiently large. By ensuring that this condition is met, the interest rate function guarantees that the equilibrium utilization U^* converges to unity as ϵ goes to zero without ever being equal to unity. In that case, the PLF interest rates converge to the market clearing rate as excess demand converges to zero without increasing excess supply.

Designing such an interest rate function ρ that guarantees $U^*(\theta) \in (1 - \epsilon, 1)$ for all θ is achieved by first *targeting* a minimal equilibrium utilization rate $1 - \epsilon$. Next, the PLF sets $\rho(1 - \epsilon) = x_\epsilon$ where x_ϵ is the (unique) value that satisfies the following equation:

$$1 - \epsilon = \theta \frac{1 - F_b(x_\epsilon)}{F_l((1 - \epsilon) \cdot x_\epsilon)} \quad (4.1)$$

Note that when setting $\rho(1 - \epsilon) = x_\epsilon$, we guarantee that $U^*(\underline{\theta}) = 1 - \epsilon$ by the construction of x_ϵ . Finally, the optimal solution sets ρ as any increasing function such that $\rho(U) \rightarrow +\infty$ as $U \rightarrow 1$.⁹ We prove that this design will ensure that $U^*(\theta) \in (1 - \epsilon, 1)$ for all $\theta \in [\underline{\theta}, +\infty)$ and use the fact that ϵ is arbitrarily chosen to guarantee that we can design ρ to ensure that all equilibrium utilization rates are arbitrarily close to 1.

We next show that Proposition 4.1 implies that the DeFi equilibrium welfare can be made arbitrarily close to the competitive equilibrium welfare through an optimally designed interest rate function. Denote by $\mathcal{W}^{DeFi}(\theta)$ and $\mathcal{W}^{CE}(\theta)$ the DeFi equilibrium and competitive welfare, respectively, when the state is θ . In that case, we denote by $\mathbb{E}[\mathcal{W}^{DeFi}]$ and $\mathbb{E}[\mathcal{W}^{CE}]$, the expected welfare in the DeFi and competitive equilibrium respectively, where the expectation is taken with respect to $\theta \sim G$.

Proposition 4.2. Expected Welfare Can Be Approximately Efficient

Suppose that $\theta \sim G[\underline{\theta}, +\infty)$ where $\underline{\theta} > 0$. For any $\delta > 0$ there exists an interest rate function ρ such that $\mathbb{E}[\mathcal{W}^{DeFi}] \geq \mathbb{E}[\mathcal{W}^{CE}] - \delta$.

When designing ρ to satisfy $U^*(\theta) \in [1 - \epsilon, 1)$ for each $\theta \in [\underline{\theta}, +\infty)$ we can ensure that the borrower and lender rates become arbitrarily close to the market clearing rate, $r^C(\theta)$, for all $\theta \in [\underline{\theta}, +\infty)$ by picking an increasingly small value of ϵ . Thus, welfare being a continuous function of interest rates implies that expected DeFi welfare can be made arbitrarily close to the competitive equilibrium welfare as interest rates converge to the market clearing rates for every level of market tightness θ .

Intuitively, as ϵ converges to zero, the approximately optimal interest rate function approaches a straight line and then becomes undefined. As mentioned above, the interest rate function $\rho(U)$ must be able to span all of the market clearing rates $\{r^C(\theta)\}_{\theta \in [0, +\infty)}$ (i.e., for all θ there must exist U such that $\rho(U) = r^C(\theta)$). However, for a strictly zero ϵ , $\rho(1)$ must

⁹Note that this is not necessary if the distribution for θ is bounded above, in which case we can choose ρ such that $\rho(1) = B$ for some sufficiently large constant B .

be equal to $r^C(\theta)$ for all θ , which is not feasible. In contrast, for any $\epsilon > 0$, it is possible to span all of the market clearing rates provided that ρ has a sufficiently low intercept and sufficiently large slope for all $U \in [1 - \epsilon, 1)$. We propose an intuitive illustration of this finding through an application using linear interest rate functions in the next section.

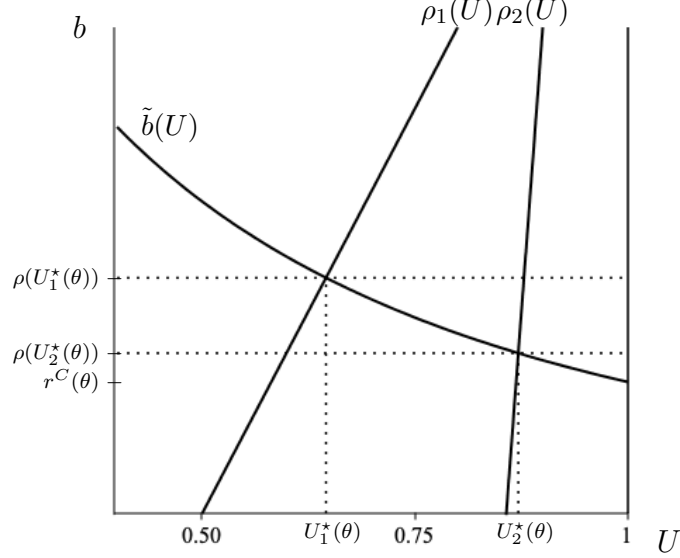


Figure 2: An illustration of the optimal interest rate design.

Graphical Representation The intuition of Proposition 4.1 is illustrated in Figure 2 where we plot two linear interest rate functions $\rho_1(U)$ and $\rho_2(U)$. As can be seen, $\rho_2(U)$ has a higher slope and lower intercept than $\rho_1(U)$. One can show that $\rho_2(U)$ generates interest rates that are closer to the market clearing rate than $\rho_1(U)$ when the market clearing condition is θ . In particular, we also plot in Figure 2 the curve $\tilde{b}(U)$, which denotes the implicit function that satisfies the following equation:

$$\theta \cdot \frac{1 - F_b(\tilde{b}(U))}{F_l(U \cdot \tilde{b}(U))} = U,$$

for any $U \in [0, 1]$. The function $\tilde{b}(U)$ represents the interest rate that ensures that the equilibrium utilization rate is equal to U in the state θ when setting the borrower interest rate equal to $\tilde{b}(U)$ and the lender interest rate equal to $U \cdot \tilde{b}(U)$. Hence, for any given interest rate function ρ , the DeFi equilibrium corresponds to the point at which $\rho(U^*(\theta)) = \tilde{b}(U^*(\theta))$. For instance in the state θ , the utilization rate under the interest rate function ρ_1 , denoted by $U_1^*(\theta)$, is given by the value where the curve $\tilde{b}(U)$ intersects $\rho_1(U)$. The same logic holds for $U_2^*(\theta)$ which is the value such that the curve $\tilde{b}(U)$ and $\rho_2(U)$ intersect. Finally, note that by construction $\tilde{b}(1) = r^C(\theta)$ which is the market clearing rate in state θ . Thus, moving from the interest rate function ρ_1 to the interest rate function ρ_2 , the equilibrium utilization rate becomes closer to the market clearing rate $r^C(\theta)$ and therefore the DeFi equilibrium becomes more efficient.

While Figure 2 illustrates the efficiency gain from increasing the slope of the interest rate function for a given θ , a crucial feature of the above results is that this efficiency gain is also realized for all other θ' . In the context of Figure 2, this result holds because the curve $\tilde{b}(U)$ shifts up for all U when $\theta' > \theta$ and shifts down for all U when $\theta' < \theta$. Thus, the high slope of ρ_2 will also help to ensure that the equilibrium utilization rates are closer to market clearing in all other states θ' . Finally, if the slope of the interest rate function ρ is sufficiently large, then the set of states θ such that $U^*(\theta) = 1$ is also minimized, which ensures that there is no spike in excess lending so that equilibrium utilization rates remain strictly less than unity for all but the largest values of θ .¹⁰

¹⁰In other words, as $\rho(1)$ increases, the probability that the credit market condition, θ , is so large that it requires a market clearing rate $r^C(\theta) \geq \rho(1)$ must go to zero and therefore any expected welfare loss arising due to excess supply (i.e., when $U^*(\theta) = 1$) must vanish to zero. More formally, as $\rho(1) \rightarrow +\infty$, $\mathbb{P}(\{\theta : r^C(\theta) \geq \rho(1)\}) \rightarrow 0$ and therefore $\mathbb{P}(\{\theta : U^*(\theta) = 1\}) \rightarrow 0$.

4.2 Linear PLF Example

In this section, we apply the results of Propositions 4.1 and 4.2 to demonstrate the optimal design of linear interest rate functions.

Example 1. Optimal Linear Interest Rate Functions Without Withdrawal Uncertainty

Consider the class of linear PLF interest rate functions of the form $\rho(U) = a + m \cdot U$ with $a \in \mathbb{R}$, $m \geq 0$ and suppose that $\theta \sim G[\underline{\theta}, \bar{\theta}]$. Then, for any sufficiently small $\epsilon > 0$ we can choose constants $a_\epsilon \in \mathbb{R}$ and $m_\epsilon \geq 0$ in order to guarantee that for all $\theta \in [\underline{\theta}, \bar{\theta}]$ the equilibrium utilization rates $U^*(\theta) \in (1 - \epsilon, 1)$. In this case, the optimal linear interest rate function chooses a_ϵ, m_ϵ to satisfy

$$\rho_\epsilon(1) = a_\epsilon + m_\epsilon = r^C(\bar{\theta}) \quad \text{and} \quad \rho(1 - \epsilon) = a_\epsilon + m_\epsilon \cdot (1 - \epsilon) = x_\epsilon, \quad (4.2)$$

where x_ϵ is chosen to solve the following equation.

$$1 - \epsilon = \theta \frac{1 - F_b(x_\epsilon)}{F_l((1 - \epsilon) \cdot x_\epsilon)}, \quad (4.3)$$

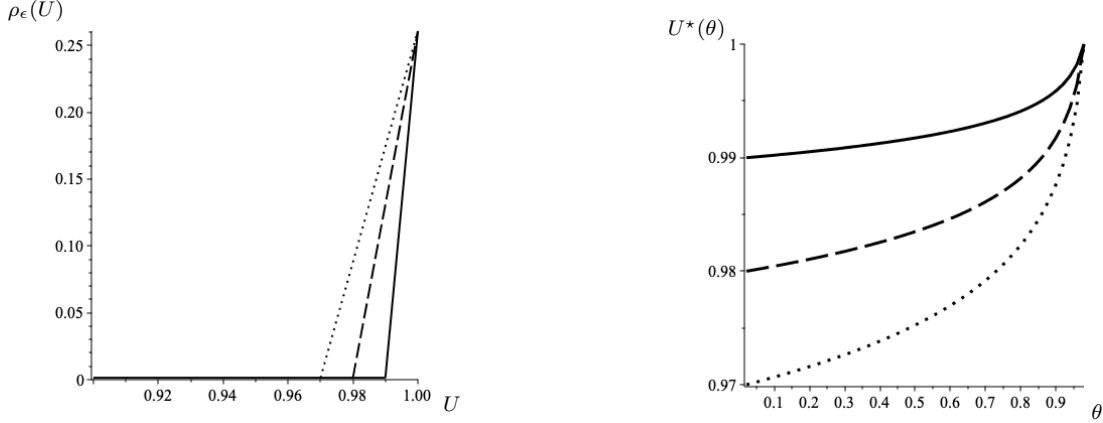
which ensures that $U^*(\theta) \geq U^*(\underline{\theta}) = 1 - \epsilon$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. Equivalently, (4.2) implies that

$$a_\epsilon = \frac{1}{\epsilon}(x_\epsilon - (1 - \epsilon)r^C(\bar{\theta})) \quad \text{and} \quad m_\epsilon = \frac{1}{\epsilon}(r^C(\bar{\theta}) - x_\epsilon).$$

As mentioned above, ρ_ϵ must satisfy the property that the intercept a_ϵ is decreasing and the slope m_ϵ is increasing as ϵ decreases.

We plot the equilibrium utilization rates as a function of θ for $\epsilon \in \{.01, .02, .03\}$ assuming $[\underline{\theta}, \bar{\theta}] = [\frac{1}{50}, \frac{49}{50}]$ and $F_l(x) = F_b(x) = 1 - e^{-15x}$. In this case, note that $r^C(\bar{\theta}) \approx .2608$ while $(x_{.01}, x_{.02}, x_{.03}) \approx (.001374, .001402, .00143)$. This implies that $(a_{.01}, a_{.02}, a_{.03}) \approx$

$(-25.68, -12.71, -8.39)$ while $(m_{.01}, m_{.02}, m_{.03}) \approx (25.94, 12.97, 8.65)$. In order to avoid negative lending rates, we could also set $\rho_\epsilon(U) = 0$ for all $U < 1 - \epsilon$ (although this is not necessary). Figure 3 (a) demonstrates the optimal design of the interest rate functions $\rho_\epsilon(U)$ for $\epsilon \in \{.01, .02, .03\}$. As can be seen from Figure 3 (b), when utilizing the optimal linear interest rate functions, all equilibrium utilization rates $U^*(\theta) \geq 1 - \epsilon$ for each $\epsilon \in \{.01, .02, .03\}$.



(a) The optimal linear interest rate functions $\rho_\epsilon(U)$ for $\epsilon = .01$ (solid), $\epsilon = .02$ (dashed), and when $\epsilon = .03$ (dotted).

(b) Equilibrium utilization rates $U^*(\theta)$ under the optimal linear $\rho_\epsilon(U)$ for $\epsilon = .01$ (solid), $\epsilon = .02$ (dashed), and when $\epsilon = .03$ (dotted).

Figure 3: Illustration of Proposition 4.1 for the optimal linear interest rate functions.

5 Optimal PLF Design With Withdrawal Shocks

In this section, we return to the full model with withdrawal shocks as introduced in Section 2. In particular, recall that the key elements of the full model are that: (i) with some probability q , a measure η of lenders need to withdraw their capital and face a cost for not being able to do so and (ii) this lending shock generates interest rate risk for borrowers as the realized utilization is different from the expected utilization at the time of borrowing decisions. Our objective in this section is to illustrate the existence of an efficiency-variance trade-off. To keep this point simple, we do so by considering linear interest rate functions

of the form $\rho(U) = m \cdot U + a$ with $m \in [0, +\infty)$ and $a \in \mathbb{R}$. In addition, we further assume for simplicity that θ has a bounded distribution by assuming that G has the support $[\underline{\theta}, \bar{\theta}]$.¹¹ We show below that with withdrawal shocks, it is no longer optimal to set an arbitrarily large slope for the interest rate function. This result is a consequence of the fact that the sensitivity — and therefore the volatility — of interest rates to the withdrawal shocks is increasing in the slope of the PLF interest rate function.

Utilization Rate Target In what follows, we make the dependence of equilibrium utilization rates on the underlying interest rate function ρ explicit by denoting the utilization rate as $U^*(\theta; \rho)$. To characterize the optimal interest rate function, we first note that if some linear interest rate function ρ generates an equilibrium utilization rate $U^*(\bar{\theta}; \rho) > 1 - \eta$, then there is a positive measure set of realizations of $\theta \in [\underline{\theta}, \bar{\theta}]$ under which the utilization rate at the start of the period $U^*(\theta; \rho) > 1 - \eta$. In this case, each lender faces a positive probability of paying the cost κ for not being able to withdraw early. Namely, recall that when the withdrawal shock occurs, a mass η of lenders wish to withdraw their funds, but only a portion $1 - U^*(\theta; \rho) < \eta$ of the funds deposited in the protocol is available. For a large enough cost of not being able to withdraw, κ , an interest rate function having this outcome in its support will not be optimal. Rather, it is always possible to design ρ such that $U^*(\bar{\theta}; \rho) \leq 1 - \eta$. For this reason, we simplify the analysis by assuming a high enough κ so that the platform acts as if it was facing an upper bound of maintaining a utilization rate $U^*(\theta; \rho) \leq 1 - \eta$ for any state of the world, $\theta \in [\underline{\theta}, \bar{\theta}]$.¹² This simplifying assumption also matches the design of utilization rate targets below unity that are set by DeFi lending

¹¹Note that by choosing an increasingly large $\bar{\theta}$, the loss in welfare from assuming a bounded $\bar{\theta}$ must necessarily go to zero as in $\mathbb{P}(\theta > \bar{\theta}) = 1 - G(\bar{\theta}) \rightarrow 0$ as $\bar{\theta} \rightarrow 0$ and welfare is bounded below by 0 for each credit market condition.

¹²Note that this restriction would be trivially met for any parameters by assuming $\kappa = +\infty$. Our result of an interior solution for the slope of the PLF is robust to instead assuming a finite κ , although at the cost of introducing an additional trade-off given that it may be optimal to violate the utilization rate target $1 - \eta$ in some states of the world so long as those states have a sufficiently low probability of occurring.

platforms in practice — such as AAVE and Compound — with $1 - \eta$ corresponding to those utilization rate targets.

Linear Design As we are focused on linear interest rate functions, in what follows, we define

$$IR := \{\rho(\cdot) : \rho(U) = a + m \cdot U, m \geq 0, a \in \mathbb{R}, U^*(\bar{\theta}; \rho) \leq 1 - \eta\},$$

as the set of linear interest rate functions $\rho(U) = a + m \cdot U$ satisfying $U^*(\bar{\theta}; \rho) \leq 1 - \eta$. This latter condition ensures that the utilization target is met for the tightest credit market condition, $\bar{\theta}$. Moreover, note that the linearity of ρ implies that $U^*(\theta; \rho) \leq U^*(\bar{\theta}; \rho)$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. Therefore any $\rho \in IR$ guarantees that the utilization rate target is always met for all credit market conditions $\theta \in [\underline{\theta}, \bar{\theta}]$.

As a first step to our analysis, note that for any given $\rho \in IR$, Equation (2.12) implies that the variance of interest rates when the credit market condition is θ , denoted by $VAR(\theta; \rho)$ is given by

$$VAR(\theta; \rho) := \iota \cdot (m \cdot U^*(\theta; \rho))^2,$$

where $\iota := (\frac{\eta}{1-\eta})^2 \cdot q \cdot (1 - q)$. In addition, we use the notation: $\mathbb{E}[\rho(\theta)] := \mathbb{E}[\rho(U)]|_{U=U^*(\theta; \rho)}$ and $\mathbb{E}[U(\theta) \cdot \rho(\theta)] := \mathbb{E}[U \cdot \rho(U)]|_{U=U^*(\theta; \rho)}$. Finally, using (2.13) and (2.14), for any $\rho \in IR$, the expected borrower and lender rates in state θ are given (respectively) by

$$\begin{aligned} \mathbb{E}[\rho(\theta)] &= a + m \cdot \left(1 + \frac{\eta}{1-\eta} \cdot q\right) \cdot U^*(\theta; \rho), \\ \mathbb{E}[U(\theta) \cdot \rho(U(\theta))] &= U^*(\theta; \rho) \cdot \left((1 - q) \cdot (a + m \cdot U^*(\theta; \rho)) + \frac{q}{1-\eta} (a + m \cdot \frac{U^*(\theta; \rho)}{1-\eta})\right). \end{aligned} \quad (5.1)$$

Welfare Optimization In what follows the platform designer chooses $\rho \in IR$ to maximize expected welfare. In particular, denoting by $\mathbb{E}[\hat{\rho}(U_t)] := \mathbb{E}[\rho(U_t)] + \zeta \cdot \text{VAR}(b_t)$ and by $\mathcal{W}^{DeFi}(\theta; \rho)$ the expected equilibrium welfare when the credit market condition is θ and the interest rate function of the PLF is $\rho \in IR$, we write:

$$\mathcal{W}^{DeFi}(\theta; \rho) = \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}^*(\theta)]}^{+\infty} (r - \mathbb{E}[\hat{\rho}^*(\theta)]) dF_b(r) + \lambda_t \cdot \int_0^{\mathbb{E}[U^*(\theta) \cdot \rho^*(\theta)]} (\mathbb{E}[U^*(\theta) \cdot \rho^*(\theta)] - r) dF_l(r),$$

which represents the sum of borrower and lender surplus.¹³ The welfare maximizing interest rate function therefore solves the following problem:

$$\max_{\rho \in IR} \int_{\underline{\theta}}^{\bar{\theta}} \mathcal{W}^{DeFi}(\theta; \rho) dG(\theta). \quad (5.2)$$

The following result states that the optimal interest rate function $\rho \in IR$ must be set to ensure that when the market condition is at its highest level, $\bar{\theta}$, the equilibrium utilization under ρ is exactly equal to the utilization target $1 - \eta$ (i.e., the optimal $\rho \in IR$ must satisfy $U^*(\bar{\theta}; \rho) = 1 - \eta$).

Proposition 5.1. Utilization Target Constraint

The optimal interest rate function $\rho(U) = a + m \cdot U \in IR$ satisfies $U^*(\bar{\theta}; \rho) = 1 - \eta$.

Proposition 5.1 states that the optimal linear interest rate function $\rho \in IR$ must satisfy the condition that $U^*(\bar{\theta}; \rho) = 1 - \eta$ and therefore for any level of market tightness $\theta \in [\underline{\theta}, \bar{\theta}]$ the PLF achieves its utilization rate target $U^*(\theta; \rho) \leq 1 - \eta$ and the utilization rate is exactly equal to the target for the highest level of market tightness $\bar{\theta}$. To prove this result, we show

¹³Note that maximizing borrower and lender surplus is equivalent to maximizing the sum of utilities given that the sum of borrower and lender utilities is equal to the sum of borrower and lender surplus plus a constant. In addition, note that this derivation of welfare assumes (as discussed in footnote 4) that in the state of the world where the withdrawal shock occurs, then the lenders who are shocked (with probability q) and withdraw receive the same utility as the lenders that are not shocked and continue lending to the platform. This assumption simplifies the analysis but is without loss when compared to the case whereby the utility of the lenders who are shocked and withdraw is exogenously specified.

that for any interest rate function $\rho \in IR$ such that $U^*(\bar{\theta}; \rho) < 1 - \eta$ there exists another interest rate function $\rho' \in IR$ that achieves strictly higher equilibrium utilization rates than ρ for each state θ while setting a strictly lower slope. Hence, ρ' generates higher equilibrium utilization rates while also ensuring a strictly lower variance of interest rates through a lower slope. In that case, such an interest rate function necessarily must produce more efficient (risk-adjusted) equilibrium interest rates and, therefore, higher expected equilibrium welfare.

Proposition 5.1 also allows us to more precisely specify the welfare maximization problem (5.2) with respect to $\rho \in IR$ by including the constraint $U^*(\bar{\theta}, \rho) = 1 - \eta$. In order to be sure that we can always satisfy this constraint, we introduce the following intermediate result:

Proposition 5.2. Satisfying the Utilization Target Constraint

For any $m \in [0, +\infty)$ denote by $\bar{x}(m)$ the value of x that solves the following equation

$$1 - \eta = \bar{\theta} \cdot \frac{1 - F_b(x + \zeta \cdot \iota \cdot (m(1 - \eta))^2)}{F_l((1 - q)(1 - \eta)(x - m \cdot q \cdot \eta) + q(x + m \cdot (1 - q) \cdot \eta))}. \quad (5.3)$$

Then, the following results hold:

- 1.) For any $m \in [0, +\infty)$ the solution $\bar{x}(m)$ exists, is unique, and $\frac{d}{dm} \bar{x}(m) < 0$.
- 2.) For any $\rho \in IR$ with slope m , if $U^*(\bar{\theta}; \rho) = 1 - \eta$ then $\mathbb{E}[\rho(U)]|_{U=1-\eta} = \bar{x}(m)$.

Proposition 5.2 demonstrates that for any slope $m \in [0, +\infty)$, the constraint $U^*(\bar{\theta}; \rho) = 1 - \eta$ can always be met, which is achieved by choosing $\rho \in IR$ to satisfy the following equation:

$$\mathbb{E}[\rho(U)]|_{U=1-\eta} = (1 - q)(a + m \cdot (1 - \eta)) + q(a + m \cdot 1) = \bar{x}(m), \quad (5.4)$$

where $\bar{x}(m)$ is the solution to (5.3), and is precisely the expected interest rate that guarantees that the equilibrium utilization is $1 - \eta$ when the state is $\bar{\theta}$ (i.e., $U^*(\bar{\theta}; \rho) = 1 - \eta$). Next, note that the constraint (5.4) allows us to derive the optimal intercept $a \in \mathbb{R}$ as a function

of the slope $m \in [0, +\infty)$. In particular, defining the function $a^*(m)$ as

$$a^*(m) := \bar{x}(m) - m \cdot (1 - (1 - q) \cdot \eta),$$

then for any m , the linear interest rate ρ with intercept $a^*(m)$ and slope m is guaranteed to satisfy (5.4) and therefore to satisfy $U^*(\bar{\theta}; \rho) = 1 - \eta$. Thus, the welfare-maximizing interest rate function must solve the following simplified problem:

$$\max_{m \in [0, +\infty)} \int_{\underline{\theta}}^{\bar{\theta}} \mathcal{W}^{DeFi}(\theta; a^*(m) + m \cdot U) dG(\theta), \quad (5.5)$$

where $a^*(m) = \bar{x}(m) - m \cdot (1 - (1 - q) \cdot \eta)$. This leads to our final result.

Proposition 5.3. *Assume that $\zeta > 0$. There exists an optimal interest rate function $\rho^* = a^* + m^* \cdot U \in IR$ whereby $a^* = \bar{x}(m^*) - m^*(1 - (1 - q) \cdot \eta)$ and $m^* \in [0, +\infty)$ is the solution to (5.5).*

Proposition 5.3 demonstrates that the introduction of withdrawal shocks creates a trade-off in the optimal slope of the interest rate function. Following a similar logic as the results from Section 4: When $\rho(U) = a^*(m) + m \cdot U$, increasing the slope of m leads to more efficient interest rates which produce higher utilization rates for all $\theta \in [\underline{\theta}, \bar{\theta}]$ when investors are not risk averse. In contrast, once investors are risk averse, a higher slope produces a higher variance of interest rates, which, once the slope m becomes sufficiently large, offsets the efficiency gains and thus lowers the equilibrium expected welfare.

We illustrate the intuition of Proposition 5.3 in Figures 4 and 5. In particular, Figure 4(a) plots the equilibrium (risk-adjusted) borrower rates and lender rates as a function of the slope of the interest rate function $\rho(U) = a^*(m) + m \cdot U$. As can be seen, as m increases, the equilibrium risk-adjusted borrower rate decreases while the lending rate increases. Such a feature ensures that equilibrium utilization rates are strictly increasing in m for all $\theta \in [\underline{\theta}, \bar{\theta}]$

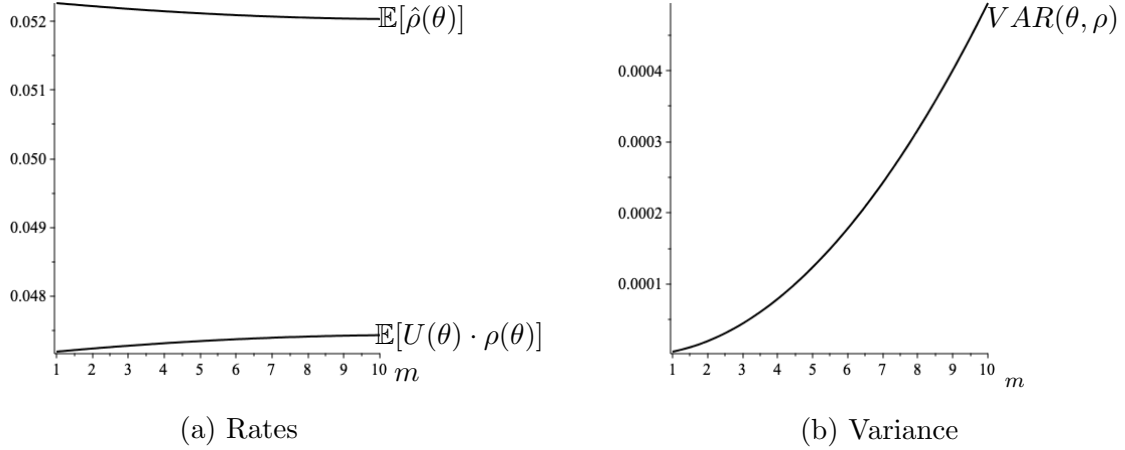


Figure 4: Equilibrium (risk-adjusted) borrower rates, lender rates, and variance as a function of the interest rate function slope, m , when $\rho(U) = a^*(m) + m \cdot U$, $F_l(x) = F_b(x) = 1 - e^{-15x}$, $\eta = .1$, $q = .01$, $\zeta = .05$, and $\theta = \bar{\theta} = 1$.

as shown in Figure 5(a). Note that without withdrawal shocks, this effect would always improve welfare (as is the case in Section 4) as it leads to lower borrower interest rates and higher lending rates, strictly improving the utility of both borrowers and lenders. In contrast, with withdrawal shocks, even though borrowing rates are decreasing, the volatility of interest rates is strictly increasing, as can be seen in Figure 4(b). Hence, while borrowers pay lower interest, they incur higher volatility. For this reason, for a sufficiently large slope $\tilde{m}$, it must be the case that borrower utility is strictly decreasing for $m > \tilde{m}$. In addition, we show that for a slope large enough, equilibrium lending rates start declining, ensuring that total welfare must also decline. We illustrate this effect in 5(b), which plots equilibrium welfare for three different states θ as a function of the slope m . As can be seen, for a sufficiently large slope, the efficiency gain from higher utilization rates is strictly offset by the welfare loss due to higher volatility. Given that for any θ we can ensure that the optimal slope is interior, the slope that maximizes expected welfare is also interior.

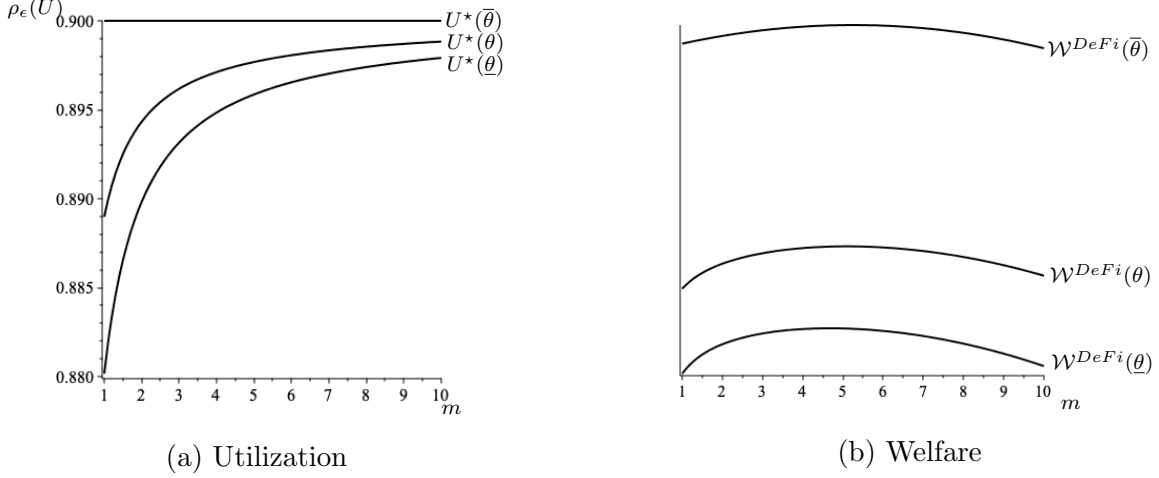


Figure 5: Equilibrium utilization rates and welfare as a function of the interest rate function slope, m , when $\rho(U) = a^*(m) + m \cdot U$, $\bar{\theta} = 1$, $\theta = .7$, $\underline{\theta} = .5$, $F_l(x) = F_b(x) = 1 - e^{-15x}$, $\eta = .1$, $q = .01$, and $\zeta = .05$.

6 Conclusion

In this paper, we develop an economic model of a DeFi lending market facilitated by a Protocol for Loanable Funds (PLF). PLFs set interest rates as a function of the observed ratio of borrowed to lent funds. We focus on the design of this interest rate setting mechanism and demonstrate the inefficiency of equilibrium outcomes in DeFi lending markets that utilize such interest rate functions. PLFs face a natural disadvantage when compared to a competitive lending market, given that technical frictions prevent the PLF from incorporating off-chain information into the platform's interest rates. Despite this informational disadvantage, we show that when there is no uncertainty in withdrawals from the platform, then it is possible to design the PLF interest rate function to ensure that the resulting DeFi equilibrium achieves a level of welfare that is arbitrarily close to first-best, independent of the underlying state of the credit market. One caveat of this design is that it requires the PLF to set its interest rate function with a large slope. With uncertainty about withdrawal demand, however, this design is no longer optimal as this uncertainty will produce large

volatility of PLF interest rates when setting a large slope, discouraging the use of the PLF by risk-averse agents. As a consequence, we demonstrate that the optimal design of the PLF interest rate function when there are withdrawal shocks must trade off the efficiency versus the volatility of PLF interest rates. Our tractable framework and its insights have several direct applications to the design of practical DeFi lending platforms and could easily be extended to consider additional forces not considered in this work, which is left to future research.

References

- Aramonte, S., S. Doerr, W. Huang, A. Schrimpf, et al. 2022. DeFi lending: intermediation without information? Tech. rep., Bank for International Settlements.
- Biais, B., C. Bisière, M. Bouvard, and C. Casamatta. 2019. The Blockchain Folk Theorem. *Review of Financial Studies* 32(5):1662–1715.
- Capponi, A., G. Iyengar, and J. Sethuraman. 2023. Decentralized Finance: Protocols, Risk, and Governance. *Foundations and Trends in Privacy and Security* 5:144–188.
- Capponi, A., and R. Jia. 2021. The Adoption of Blockchain-based Decentralized Exchanges. *Columbia University Working Paper* .
- Carré, S., and F. Gabriel. 2022. Security and Efficiency in DeFi Lending. *Working Paper* 25:3389–3421.
- Chaudhary, A., R. Kozhan, and G. Vishwanath-Natraj. 2023. Interest Rate Parity in Decentralized Finance. *working paper* .
- Chiu, J., E. Ozdenoren, K. Yuan, and S. Zhang. 2022. On the Fragility of DeFi Lending. *Available at SSRN 4328481* .
- Cong, L. W., Y. Li, and N. Wang. 2021. Tokenomics: Dynamic Adoption and Valuation. *Review of Financial Studies* 34(3):1105–1155.
- d’Avernas, A., V. Maurin, and Q. Vandeweyer. 2022. Can Stablecoins be Stable? *Working Paper* .
- Easley, D., M. O’Hara, and S. Basu. 2019. From Mining to Markets: The Evolution of Bitcoin Transaction Fees. *Journal of Financial Economics* 134(1):91–109.

- Hasbrouck, J., T. Rivera, and F. Saleh. 2023. The Need for Fees at a DEX: How Increases in Fees Can Increase DEX Trading Volume. *Working Paper* .
- John, K., L. Kogan, and F. Saleh. 2023. Smart Contracts and Decentralized Finance. *Annual Review of Financial Economics* Forthcoming.
- Lehar, A., and C. Parlour. 2024. Decentralized Exchange: The Uniswap Automated Market Maker. *Journal of Finance* Forthcoming.
- Lehar, A., and C. A. Parlour. 2022. Systemic fragility in decentralized markets. *Available at SSRN* .
- Li, Y., and S. Mayer. 2022. Money Creation in Decentralized Finance: A Dynamic Model of Stablecoin and Crypto Shadow Banking. *Working Paper* .
- Mayer, S. 2022. Token-Based Platforms and Speculators. *Working Paper* .
- Mueller, P. 2022. DeFi Leveraged Trading: Inequitable Costs of Decentralization. *Available at SSRN 4241356* .
- Park, A. 2021. The Conceptual Flaws of Constant Product Automated Market Making. *University of Toronto Working Paper* .
- Park, A., and J. Stinner. 2023. Phantom Liquidity in Decentralized Lending. *Working Paper* 25:3389–3421.
- Saleh, F. 2021. Blockchain Without Waste: Proof-of-Stake. *Review of Financial Studies* 34(3):1156–1190.

Appendices

A Proofs

A.1 Proof of Proposition 3.3

Proof. To prove existence of U^* , first note that we have assumed that $\rho(U) > 0$ for some $U \in [0, 1)$. We will proceed in two cases:

Case 1: $\rho(U) > 0$ for all $U \in [0, 1]$: In this case we know that

$$\frac{\partial}{\partial U} [\min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\}] \leq 0, \quad (\text{A.1})$$

which comes directly from the fact that

$$\frac{\partial}{\partial U} [x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}] = -x \cdot \frac{F'_b(U)\rho'(U)F_l(U\rho(U)) + F'_l(U\rho(U)) \cdot (\rho(U) + U\rho'(U)) \cdot (1 - F_b(\rho(U)))}{(F_l(U \cdot \rho(U)))^2} \leq 0.$$

Further, note that

$$0 < \lim_{U \rightarrow 0} \min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} = 1 \quad \text{and} \quad 1 \geq \min\{x \cdot \frac{1 - F_b(\rho(1))}{F_l(\rho(1))}, 1\},$$

and therefore, for all $x \in [0, +\infty)$, there must exist U such that

$$U = \min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\}. \quad (\text{A.2})$$

Case 2: There exists U_0 such that $\rho(U_0) = 0$: In this case, we know that $\rho(U) \leq 0$ for all $U \leq U_0$ and $\rho(U) \geq 0$ for all $U \geq U_0$. Further, the existence of U such that $\rho(U) > 0$ implies that there exists $U_0 \in [0, 1]$ such that $\rho(U) > 0$ and weakly increasing for all $U \in (U_0, 1]$

while $\rho(U_0) = 0$.

In this case, note that $U < U_0$ implies that $U \cdot \rho(U) \leq 0$. Next, using the convention that $F_l(x) = F_b(x) = 0$ for all $x \leq 0$, then using the convention that $U = 1$ whenever $\mathcal{S}(l) = 0$ and $\mathcal{D}(b) > 0$ then $\min\{x \cdot \frac{1-F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} = 1$ whenever $U \cdot \rho(U) \leq 0$. In that case, we know that $U \cdot \rho(U) > 0$ and weakly increasing for $U \in (U_0, 1]$ and $U_0 \cdot \rho(U_0) = 0$. Hence,

$$\min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\},$$

is equal to 1 for all $U \in [0, U_0]$ and weakly decreasing for all $U \in (U_0, 1]$. Therefore, it must be the case that

$$0 < \lim_{U \rightarrow^+ U_0} \min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} = 1 \quad \text{and} \quad 1 \geq \min\{x \cdot \frac{1 - F_b(\rho(1))}{F_l(\rho(1))}, 1\},$$

which combined with the fact that $\min\{x \cdot \frac{1-F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} = 1 > U$ whenever $U \cdot \rho(U) \leq 0$ implies that for all $x > 0$ there must exist a solution $U^*(x)$ to (A.2). Further, $U^*(x) \in (U_0, 1]$.

To prove uniqueness, suppose by contradiction that for some $x \in [0, +\infty)$ there exists $U \neq U'$ such that

$$U = \min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} \quad \text{and} \quad U' = \min\{x \cdot \frac{1 - F_b(\rho(U'))}{F_l(U' \cdot \rho(U'))}, 1\}.$$

Further, assume without loss of generality that $U > U'$. Then, $U - U' > 0$ which implies that

$$\min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} - \min\{x \cdot \frac{1 - F_b(\rho(U'))}{F_l(U' \cdot \rho(U'))}, 1\} > 0.$$

Yet, (A.1) implies that

$$\min\{x \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}, 1\} - \min\{x \cdot \frac{1 - F_b(\rho(U'))}{F_l(U' \cdot \rho(U'))}, 1\} \leq 0,$$

whenever $U > U'$, a contradiction. □

A.2 Proof of Proposition 3.2

Proof. Let $z(U, \theta)$ be defined as

$$z(U, \theta) = \theta \cdot \frac{1 - F_b(\rho(U))}{F_l(U \cdot \rho(U))}.$$

Then, consider $\theta < \theta'$ and denote by

$$U = \min\{\theta \cdot z(U, \theta), 1\} \quad \text{and} \quad U' = \min\{\theta' \cdot z(U', \theta'), 1\},$$

with the existence and uniqueness of U and U' guaranteed by Proposition 3.3. We further we can see that $\frac{d}{d\theta}z(U, \theta) > 0$ and we know from the proof of 3.3 that $\frac{d}{dU}z(U, \theta) \leq 0$ for all $U \in [0, 1]$ and all $\theta \in [0, +\infty)$. Now, by contradiction suppose that $U > U'$. Then,

$$U' = \min\{\theta' \cdot z(U', \theta'), 1\} \geq \min\{\theta \cdot z(U', \theta), 1\} \geq \min\{\theta \cdot z(U, \theta), 1\} = U,$$

where the last inequality comes from the fact that $U > U'$ implies that $z(U', \theta) \geq z(U, \theta)$. This presents a contradiction to the fact that we have assumed $U > U'$. Therefore, we have shown that $\theta_t < \theta_{t'}$ implies that $U_t^* \leq U_{t'}^*$.

Finally, suppose that $U < 1$ but $U = U'$. Then,

$$U = \theta \cdot z(U, \theta) < \min\{\theta' \cdot z(U, \theta'), 1\} = \min\{\theta' \cdot z(U', \theta'), 1\} = U',$$

where the first inequality comes from the fact that $\theta \cdot z(U, \theta) < 1$ by assumption and $\theta < \theta'$. Again, this presents a contradiction given that we have assumed that $U = U'$.

In order to prove that $b_t^* \leq b_{t'}^*$ and $l_t^* \leq l_{t'}^*$ whenever $\theta_t < \theta_{t'}$ we note that $b_t^* = \rho(U_t^*)$

and $l_t^* = U_t^* \cdot \rho(U_t^*)$. Therefore, noting that $\rho'(U) \geq 0$ and that it must be the case that $\frac{d}{dU}[U \cdot \rho(U)] = U \cdot \rho'(U) + \rho(U) > 0$ for the equilibrium utilization U (as demonstrated in the proof of Proposition 3.3) then implies that whenever $U_t^* \leq U_{t'}^*$ then $b_t^* \leq b_{t'}^*$ and $l_t^* \leq l_{t'}^*$ with all inequalities strict whenever ρ is strictly increasing and $U_t^* < U_{t'}^*$ which we have shown is the case whenever $U_t^* < 1$. $\square$

A.3 Proof of Proposition 3.3

Proof. The existence of r^C comes immediately from the fact that $F_l'(r) > 0$ and $F_b'(r) > 0$ for all $r \in [0, +\infty)$ and for any $x \in (0, +\infty)$

$$F_l(0) = 0 \leq x(1 - F_b(0)) = x \quad \text{and} \quad \lim_{r \rightarrow +\infty} F_l(r) = 1 > \lim_{r \rightarrow +\infty} x(1 - F_b(r)) = 0.$$

Uniqueness comes from the fact that $F_l(r)$ is strictly increasing while $x(1 - F_b(r))$ is strictly decreasing for $x \in (0, +\infty)$. Existence implies that r^C is onto while uniqueness implies that r^C is one-to-one. $\square$

A.4 Proof of Proposition 3.4

Proof. For any $\lambda, \mu \neq 0$, denote by $\mathcal{D}(r^C(\frac{\mu}{\lambda})) = \mu(1 - F_b(r^C(\frac{\mu}{\lambda})))$ and $\mathcal{S}(r^C(\frac{\mu}{\lambda})) = F_l(\frac{\mu}{\lambda})$. We will show that if λ and μ are continuously distributed then for any fixed $\alpha > 0$:

$$|\{(\lambda, \mu) : \alpha = \mathcal{S}(r^C(\frac{\mu}{\lambda})) = \mathcal{D}(r^C(\frac{\mu}{\lambda}))\}| = +\infty.$$

In particular, suppose that $\mathcal{D}(r^C(\frac{\mu}{\lambda})) = \mu(1 - F_b(r^C(\frac{\mu}{\lambda}))) = \alpha$. In order to prove our claim, we will show that for all $\mu' \neq \mu$ there exists $\lambda' \neq \lambda$ such that $\mathcal{S}(r^C(\frac{\mu'}{\lambda'})) = \mathcal{D}(r^C(\frac{\mu'}{\lambda'})) = \alpha$. In order to do so, first take any $\mu' \neq \mu$ but sufficiently close so that $|\mu' - \mu| < \epsilon$ for some $\epsilon > 0$ but small. Then, using the fact that $r^C(\frac{\mu}{\lambda})$ is continuous, strictly increasing in μ , strictly

decreasing in λ , and unbounded we know that if

$$\mu'(1 - F_b(r^C(\frac{\mu'}{\lambda}))) > \alpha,$$

then $\lim_{\lambda \rightarrow 0} r^C(\frac{\mu'}{\lambda}) = +\infty$ implies that there must exist $\lambda' < \lambda$ such that

$$\mu'(1 - F_b(r^C(\frac{\mu'}{\lambda'}))) = \alpha.$$

In contrast, if

$$\mu'(1 - F_b(r^C(\frac{\mu'}{\lambda}))) < \alpha,$$

then $\lim_{\lambda \rightarrow +\infty} r^C(\frac{\mu'}{\lambda}) = 0$ implies that there must exist ϵ sufficiently small such that $|m' - m| < \epsilon$ ensures that there exists $\lambda'' > \lambda$ such that

$$\mu'(1 - F_b(r^C(\frac{\mu'}{\lambda''}))) = \alpha.$$

Otherwise it must be the case that $\mu' < \alpha$ for all μ' such that $0 < |\mu' - \mu| < \epsilon$. Yet, $\mu \neq 0$ implies that $1 - F_b(r^C(\frac{\mu}{\lambda})) < 1$ and therefore $\mu' < \alpha$ for all μ' such that $0 < |\mu' - \mu| < \epsilon$ contradicts the fact that $\mu > \mu \cdot (1 - F_b(r^C(\frac{\mu}{\lambda}))) = \alpha$.

Using this result, we know that for a fixed $\alpha = \mathcal{D}(r^C(\frac{\mu}{\lambda}))$ there exists infinitely many unique pairs (λ', μ') such that $\alpha = \mathcal{D}(r^C(\frac{\mu}{\lambda})) = \mathcal{S}(r^C(\frac{\mu}{\lambda})) = \mathcal{D}(r^C(\frac{\mu'}{\lambda'})) = \mathcal{S}(r^C(\frac{\mu'}{\lambda'}))$. Further, if $\mathcal{D}(r^C(\frac{\mu'}{\lambda'})) = \mathcal{D}(r^C(\frac{\mu}{\lambda}))$ then it must be the case that

$$\mu(1 - F_b(r^C(\frac{\mu}{\lambda}))) = \mu'(1 - F_b(r^C(\frac{\mu'}{\lambda'}))),$$

which implies that if $\mu' > \mu$ then $\frac{\mu'}{\lambda'} > \frac{\mu}{\lambda}$ and therefore $r^C(\frac{\mu'}{\lambda'}) > r^C(\frac{\mu}{\lambda})$. Similarly, if $\mu > \mu'$ then it must be the case that $\frac{\mu'}{\lambda'} < \frac{\mu}{\lambda}$ and therefore $r^C(\frac{\mu'}{\lambda'}) < r^C(\frac{\mu}{\lambda})$. This implies that there are an infinite number of market conditions (μ, λ) that generate realized supply and demand

equal to α at the market clearing rate, with each pair (μ, λ) having a unique market clearing rate $r^C(\frac{\mu}{\lambda})$. Hence, when observing $\mathcal{S}_t = \mathcal{D}_t = \alpha$ the interest rate function cannot possibly set $\rho(\alpha, \alpha) = r^C(\frac{\mu}{\lambda})$ without precisely knowing what the true market condition (μ, λ) is. Further, if $\rho(\alpha, \alpha)$ is arbitrarily specified then this implies that there is a unique pair (μ, λ) such that $\rho(\alpha, \alpha) = r^C(\frac{\mu}{\lambda})$. Namely, while multiple pairs (μ, λ) can generate the same market clearing interest rate, we have shown above that no two such pairs can generate the same realized demand at that market clearing rate. Hence, for any interest rate function $\rho(\mathcal{S}_t, \mathcal{D}_t)$, it must be the case that $Pr(\rho(\alpha, \alpha) = r^C(\frac{\mu}{\lambda})) = 0$ for all $\alpha > 0$. $\square$

A.5 Proof of Proposition 3.5

Proof. Denote by $\mathcal{S}(l_t) = F_l(l_t)$ and $\mathcal{D}(b_t) = 1 - F_b(b_t)$. We first note that

$$\mathcal{W}_t^{DeFi} = \mu_t \cdot (1 - F_b(b_t)) \cdot \min\left\{\frac{\mathcal{S}(l_t)}{\mathcal{D}(b_t)}, 1\right\} \cdot (\mathbb{E}[r_b | r_b \geq b_t] - b_t) + \lambda_t \cdot (F_l(l_t) \cdot l_t + (1 - F_l(l_t)) \cdot \mathbb{E}[r_l | r_l \geq l_t]),$$

and

$$\mathcal{W}_t^{CE} = \mu_t \cdot (1 - F_b(r_t^C)) \cdot (\mathbb{E}[r_b | r_b \geq r_t^C] - r_t^C) + \lambda_t \cdot (F_l(r_t^C) \cdot r_t^C + (1 - F_l(r_t^C)) \cdot \mathbb{E}[r_l | r_l \geq r_t^C]).$$

Further, using the fact that r_t^C guarantees that supply equals demand for all θ_t implies $\mu_t \cdot (1 - F_b(r_t^C)) = \lambda_t \cdot F_l(r_t^C)$ and therefore

$$\mathcal{W}_t^{CE} = \mu_t \int_{r_t^C}^{+\infty} r dF_b(r) + \lambda_t \int_{r_t^C}^{+\infty} r dF_l(r),$$

for all θ_t .

Now, let $\tilde{\theta} = (r^C)^{-1}(\rho(1))$ guaranteed to exist given that r^C is a bijection. Whenever $\theta_t = \tilde{\theta}$ then $l_t^* = b_t^* = r_t^C$ so that the DeFi market supply is equal to demand and therefore

$\mathcal{W}_t^{DeFi} = \mathcal{W}_t^{CE}$. What will show is that whenever $\theta_t \neq \tilde{\theta}$ then $\mathcal{W}_t^{DeFi} < \mathcal{W}_t^{CE}$. We proceed with two cases.

Case 1: $\theta_t > \tilde{\theta}$: By construction, note that $\theta_t > \tilde{\theta}$ implies that $r_t^C > \rho(1)$. Further, given Proposition 3.2 we know that U_t^* is weakly increasing in θ_t but cannot exceed one. Therefore, it must be the case that $U_t^* = 1$ and $b_t^* = l_t^* = \rho(1)$ whenever $\theta_t > \tilde{\theta}$. Once this is the case, we know that $\theta_t > \tilde{\theta}$ implies that $\mathcal{D}(r_t^C) = \mathcal{S}(r_t^C)$ while $\mathcal{D}(b_t^*) > \mathcal{S}(l_t^*)$. Therefore, whenever $\theta_t > \tilde{\theta}$ then using the fact that

$$\mu_t(1 - F_b(b_t)) \cdot \frac{\mathcal{S}(l_t)}{\mathcal{D}(b_t)} = \lambda_t \cdot F_l(l_t),$$

we can see that in this case,

$$\mathcal{W}_t^{DeFi} = \frac{\lambda_t \cdot F_l(\rho(1))}{1 - F_b(\rho(1))} \cdot \int_{\rho(1)}^{+\infty} r dF_b(r) + \lambda_t \int_{\rho(1)}^{+\infty} r dF_l(r).$$

Furthermore, denote by

$$\Phi(\rho) := \frac{\lambda_t \cdot F_l(\rho)}{1 - F_b(\rho)} \cdot \int_{\rho}^{+\infty} r dF_b(r) + \lambda_t \int_{\rho}^{+\infty} r dF_l(r),$$

then

$$\frac{d}{d\rho} \Phi(\rho) = \frac{\lambda_t(F_l'(\rho)(1 - F_b(\rho)) + F_b'(\rho)F_l(\rho))}{(1 - F_b(\rho))^2} \int_{\rho}^{+\infty} r dF_b(r) - \lambda_t \left(\frac{F_l(\rho)}{1 - F_b(\rho)} F_b'(\rho) \cdot \rho + F_l'(\rho) \cdot \rho \right) >$$

$$\frac{\lambda_t(F_l'(\rho)(1 - F_b(\rho)) + F_b'(\rho)F_l(\rho))}{(1 - F_b(\rho))^2} \cdot (1 - F_b(\rho)) \cdot \rho - \lambda_t \left(\frac{F_l(\rho)}{1 - F_b(\rho)} F_b'(\rho) \cdot \rho + F_l'(\rho) \cdot \rho \right) = 0.$$

Therefore, given that $r_t^C > \rho(1)$ and $\Phi(\rho)$ is increasing in ρ , then recalling that r_t^C clears the

market so that $\lambda_t F_l(r_t^C) = \mu_t(1 - F_b(r_t^C))$ implies

$$\mathcal{W}_t^{DeFi} < \frac{\lambda_t \cdot F_l(r_t^C)}{1 - F_b(r_t^C)} \cdot \int_{r_t^C}^{+\infty} r dF_b(r) + \lambda_t \int_{r_t^C}^{+\infty} r dF_l(r) = \mu_t \int_{r_t^C}^{+\infty} r dF_b(r) + \lambda_t \int_{r_t^C}^{+\infty} r dF_l(r) = \mathcal{W}_t^{CE}.$$

Case 2: $\theta_t \leq \tilde{\theta}$: In this case, we will prove that setting $l_t = b_t = r_t^C$ maximizes welfare subject to the constraint $\mathcal{S}(l_t) \geq \mathcal{D}(b_t)$ which must hold whenever $\theta_t \leq \tilde{\theta}$. In particular, when $\theta_t \leq \tilde{\theta}$ then for any tuple (l_t, b_t) , welfare is given by:

$$\mathcal{W}(l_t, b_t) = \mu_t \int_{b_t}^{\infty} r dF_b(r) - \mu_t \int_{b_t}^{\infty} b_t dF_b(r) + \lambda_t \int_0^{l_t} l_t dF_l(r) + \lambda_t \int_{l_t}^{\infty} r dF_l(r).$$

Further note that,

$$\mathcal{W}(l_t, b_t) - \lambda_t \int_0^{\infty} r dF_l(r) = \mu_t \cdot \int_{b_t}^{\infty} (r - b_t) dF_b(r) + \lambda_t \cdot \int_0^{l_t} (l_t - r) dF_l(r).$$

Then it can be seen that maximizing welfare with respect to (b_t, l_t) is equivalent to solving the following optimization problem:

$$\max_{l_t, b_t} \mu_t \cdot \int_{b_t}^{\infty} (r - b_t) dF_b(r) + \lambda_t \cdot \int_0^{l_t} (l_t - r) dF_l(r),$$

s.t. $\lambda_t \cdot l_t \cdot F_l(l_t) = \mu_t \cdot b_t \cdot (1 - F_b(b_t))$, $\mathcal{S}(l_t) \geq \mathcal{D}(b_t)$ where the first constraint arises due to the fact that all interest paid by the borrowers is passed through to the lenders. Note that combining the first constraint with the second implies that $l_t \leq b_t$. Further, using the first constraint, this optimization problem is equivalent to

$$\max_{l_t, b_t} \mu_t \cdot \int_{b_t}^{\infty} r \cdot dF_b(r) - \lambda_t \cdot \int_0^{l_t} r \cdot dF_l(r), \quad (\text{A.3})$$

s.t. $\lambda_t \cdot l_t \cdot F_l(l_t) = \mu_t \cdot b_t \cdot (1 - F_b(b_t))$, $\mathcal{S}(l_t) \geq \mathcal{D}(b_t)$. Next note that

$$\frac{d}{db_t} [\mu_t \cdot \int_{b_t}^{\infty} r \cdot dF_b(r) - \lambda_t \cdot \int_0^{l_t} r \cdot dF_l(r)] = -\mu_t b_t F'_b(b_t) < 0,$$

and therefore letting l'_t and b'_t denote the solution to (A.3), then it must be the case that $l'_t = b'_t = r$. Further, whenever $l'_t = b'_t = r$ then the pass through condition implies $\lambda_t \cdot r \cdot F_l(r) = \mu_t \cdot r \cdot (1 - F_b(r))$ which can only be the case if $r = 0$ or $r = r^C$. Finally, we note that if $r = 0$ then $\mathcal{D}(0) = \mu_t > 0 = \mathcal{S}(0)$ which cannot be the case when $\theta_t \leq \tilde{\theta}$ and therefore welfare is optimized at $l'_t = b'_t = r^C$. Finally, noting that $\theta_t < \tilde{\theta}$ implies that $\mathcal{S}(l_t^*) > \mathcal{D}(b_t^*)$ so that $l_t^* = \frac{\mathcal{D}(b_t^*)}{\mathcal{S}(l_t^*)} \cdot b_t^* < b_t^*$, and therefore either $l_t^* \neq r_t^C$, $b_t^* \neq r_t^C$, or both and must be the case that $\mathcal{W}_t^{DeFi} < \mathcal{W}_t^{CE}$.

□

A.6 Proof of Proposition 3.6

Proof. We have shown that r^C is a bijection and therefore its inverse $(r^C)^{-1}$ exists. Now, let $\tilde{\theta} = (r^C)^{-1}(\rho(1))$. By construction, when $\theta_t = \tilde{\theta}$ then $U_t^* = 1$ and $b_t^* = l_t^* = \rho(1) = r_t^C$ therefore, $\mathcal{W}_t^{CE} = \mathcal{W}_t^{DeFi}$. Furthermore, given that r^C is a bijection then it must be the case that $\tilde{\theta}$ is the unique value such that $\mu_t(1 - F_b(\rho(1))) = \lambda_t F_l(\rho(1))$ and therefore by Proposition 3.3 the unique value of $\tilde{\theta}$ such that $\mathcal{W}_t^{CE} = \mathcal{W}_t^{DeFi}$.

□

A.7 Proof of Proposition 3.7

Proof. Note that $\theta_t > \tilde{\theta}$ implies that $b_t^* = l_t^* = \rho(1)$ due to the fact that $b_t^* \leq \rho(1)$, $l_t^* \leq \rho(1)$, $b_t^* = l_t^* = \rho(1)$ when $\theta_t = \tilde{\theta}$, and b_t^* and l_t^* are weakly increasing in θ_t by Proposition 3.2. Further, we know that r_t^C is strictly increasing in θ_t and therefore $r_t^C > \rho(1) = b_t^* = l_t^*$ whenever $\theta_t > \tilde{\theta}$.

We know that whenever $\theta_t < \tilde{\theta}$, then $\mu_t(1 - F_b((b_t^*))) < \lambda_t F_l(l_t^*)$ so that $U_t^* < 1$ and therefore $l_t^* < b_t^*$. In that case, we know

$$\frac{F_l(r_t^C)}{1 - F_b(r_t^C)} = \theta_t < \frac{F_l(l_t^*)}{1 - F_b(b_t^*)} < \frac{F_l(b_t^*)}{1 - F_b(b_t^*)},$$

and therefore $b_t^* > r_t^C$. The ambiguity of the relationship between l_t^* and r_t^C comes from the fact that $b_t^* = \rho(U_t^*) > r_t^C$ does not necessarily imply any relationship between $U_t^* \rho(U_t^*)$ and r_t^C . $\square$

A.8 Proof of Proposition 4.1

Proof. We proceed by construction. More explicitly, for any $\epsilon > 0$, we construct an interest rate function ρ_ϵ such that the equilibrium utilization rate arising from using this function as the PLF interest rate function (i.e., $\rho = \rho_\epsilon$) satisfies $U^*(\theta) \in (1 - \epsilon, 1)$ for all $\theta \in [\underline{\theta}, +\infty)$. We construct ρ_ϵ explicitly as follows:

$$\rho_\epsilon(U) = \frac{r^C(\underline{\theta}) + \delta_\epsilon}{1 - U} \cdot \epsilon, \quad (\text{A.4})$$

where δ_ϵ is defined as the unique solution to the following equation:

$$1 - \epsilon = \underline{\theta} \frac{1 - F_b(r^C(\underline{\theta}) + \delta_\epsilon)}{F_l((1 - \epsilon) \cdot (r^C(\underline{\theta}) + \delta_\epsilon))}. \quad (\text{A.5})$$

Note that δ_ϵ is always uniquely well-defined because the right hand side of Equation (A.5) is larger than $1 - \epsilon$ at $\delta_\epsilon = 0$ and strictly decreases to 0 as $\delta_\epsilon \rightarrow +\infty$.¹⁴

To establish the stated result, we must prove that the equilibrium PLF Utilization rate, $U^*(\theta)$, implied by $\rho = \rho_\epsilon$ satisfies $U^*(\theta) \in (1 - \epsilon, 1)$. To demonstrate this result, we define:

$$\Delta_\epsilon(x, \theta) = x - \min\left\{ \theta \cdot \frac{1 - F_b(\rho_\epsilon(x))}{F_l(x \cdot \rho_\epsilon(x))}, 1 \right\}. \quad (\text{A.6})$$

Note that $\Delta_\epsilon(1 - \epsilon, \theta) < \Delta_\epsilon(1 - \epsilon, \underline{\theta}) = 0$, that $\lim_{x \rightarrow 1^-} \Delta_\epsilon(x, \theta) = 1$ and that $\Delta_\epsilon(x, \theta)$ is strictly increasing in x . In turn, continuity of Δ_ϵ in x implies that, for an arbitrary θ , there exists a unique solution $x_\epsilon^*(\theta)$ to $\Delta_\epsilon(x_\epsilon^*(\theta), \theta) = 0$ such that $x_\epsilon^*(\theta) \in (1 - \epsilon, 1)$. Then, from Proposition 3.3, recall that $U^*(\theta)$ is the unique solution to $U^*(\theta) = \min\left\{ \theta \cdot \frac{1 - F_b(\rho_\epsilon(U^*(\theta)))}{F_l(U^*(\theta) \cdot \rho_\epsilon(U^*(\theta)))}, 1 \right\}$ so that $U^*(\theta) = \min\left\{ \theta \cdot \frac{1 - F_b(\rho_\epsilon(U^*(\theta)))}{F_l(U^*(\theta) \cdot \rho_\epsilon(U^*(\theta)))}, 1 \right\} \Leftrightarrow \Delta_\epsilon(U^*(\theta), \theta) = 0$ implies $U^*(\theta) = x_\epsilon^*(\theta) \in (1 - \epsilon, 1)$ as desired.

□

A.9 Proof of Proposition 4.2

Proof. We apply Proposition 4.1 to generate a sequence of PLF interest rate functions, $\{\rho_{\epsilon_n}(U)\}_{n=1}^\infty$, where $\epsilon_n = \frac{1}{n}$ so that the associated utilization rate, $U_n^*(\theta)$, satisfies $U_n^*(\theta) \in (1 - \frac{1}{n}, 1)$. In turn, we demonstrate that $\lim_{n \rightarrow +\infty} \mathbb{E}[\mathcal{W}_{t,n}^{DeFi}] = \mathbb{E}[\mathcal{W}_t^{CE}]$ where $\mathcal{W}_{t,n}^{DeFi}$ refers to the realized welfare from a PLF when the interest rate function is ρ_{ϵ_n} .

Note that Proposition 3.5 implies $\sup_{n \in \mathbb{N}} |\mathcal{W}_{t,n}^{DeFi}| \leq |\mathcal{W}_t^{CE}|$. In turn, since $\mathbb{E}[|\mathcal{W}_t^{CE}|] \leq$

¹⁴The fact that the right hand side is larger than $1 - \epsilon$ at $\delta_\epsilon = 0$ arises due to the definition of $r^C(\underline{\theta})$. More explicitly, as per Equation (3.7), $F_l(r^C(\underline{\theta})) = \underline{\theta} \cdot (1 - F_b(r^C(\underline{\theta})))$ which implies $(1 - \epsilon) \cdot F_l((1 - \epsilon) \cdot r^C(\underline{\theta})) < \underline{\theta} \cdot (1 - F_b(r^C(\underline{\theta})))$ and thus $1 - \epsilon < \underline{\theta} \cdot \frac{1 - F_b(r^C(\underline{\theta}))}{F_l((1 - \epsilon) \cdot (r^C(\underline{\theta})))}$.

$\mathbb{E}[|\mu_t|] \int_0^{+\infty} r \, dF_b(r) + \mathbb{E}[|\lambda_t|] \int_0^{+\infty} r \, dF_l(r) < +\infty$, we apply the Dominated Convergence Theorem to yield:

$$\lim_{n \rightarrow +\infty} \mathbb{E}[\mathcal{W}_{t,n}^{DeFi}] = \mathbb{E}[\lim_{n \rightarrow +\infty} \mathcal{W}_{t,n}^{DeFi}]. \quad (\text{A.7})$$

Then, since welfare is a continuous function of the borrowing and lending rates (see Equations 3.5 and 3.6), it is sufficient for this proof to demonstrate that the lending and borrowing rates at the PLF converge to the competitive equilibrium interest rate because then $\lim_{n \rightarrow +\infty} \mathcal{W}_{t,n}^{DeFi} = \mathcal{W}_t^{CE}$ which, via Equation (A.7), further implies $\lim_{n \rightarrow +\infty} \mathbb{E}[\mathcal{W}_{t,n}^{DeFi}] = \mathbb{E}[\mathcal{W}_t^{CE}]$ as desired.

To demonstrate that the lending and borrowing rates at the PLF converge to the competitive equilibrium rate, note that, when the PLF interest rate function is ρ_{ϵ_n} , then the equilibrium PLF borrowing rate in period t , $b_n^*(\theta_t)$, must satisfy:

$$U_n^*(\theta_t) = \theta_t \cdot \frac{1 - F_b(b_n^*(\theta_t))}{F_l(U_n^*(\theta_t) \cdot b_n^*(\theta_t))}. \quad (\text{A.8})$$

Recall that $U_n^*(\theta_t) \in (1 - \frac{1}{n}, 1)$ implying that $\lim_{n \rightarrow +\infty} U_n^*(\theta_t) = 1$. In turn, taking limits as $n \rightarrow +\infty$ on both sides of Equation (A.8) and using continuity of F_b and F_l implies:

$$1 = \theta_t \cdot \frac{1 - F_b(\lim_{n \rightarrow +\infty} b_n^*(\theta_t))}{F_l(\lim_{n \rightarrow +\infty} b_n^*(\theta_t))} \Leftrightarrow F_l(\lim_{n \rightarrow +\infty} b_n^*(\theta_t)) = \theta_t (1 - F_b(\lim_{n \rightarrow +\infty} b_n^*(\theta_t))). \quad (\text{A.9})$$

Equation (A.9) then implies $\lim_{n \rightarrow +\infty} b_n^*(\theta_t) = r^C(\theta_t)$ because this equation uniquely defines the competitive equilibrium interest rate as per Proposition 3.3. Moreover, recall that the PLF lending rate, $l_n^*(\theta_t)$, is given explicitly by $l_n^*(\theta_t) = U_n^*(\theta_t) \cdot b_n^*(\theta_t)$ so that $\lim_{n \rightarrow +\infty} l_n^*(\theta_t) = \lim_{n \rightarrow +\infty} U_n^*(\theta_t) \cdot \lim_{n \rightarrow +\infty} b_n^*(\theta_t) = 1 \cdot r^C(\theta_t) = r^C(\theta_t)$. Finally, $\lim_{n \rightarrow +\infty} (b_n^*(\theta_t), l_n^*(\theta_t)) = (r^C(\theta_t), r^C(\theta_t))$ implies $\lim_{n \rightarrow +\infty} \mathcal{W}_{t,n}^{DeFi} = \mathcal{W}_t^{CE}$ which implies $\lim_{n \rightarrow +\infty} \mathbb{E}[\mathcal{W}_{t,n}^{DeFi}] = \mathbb{E}[\mathcal{W}_t^{CE}]$ as per Equation (A.7), thereby completing the proof. □

A.10 Proof of Proposition 5.1

Proof. In order to prove the result we will show that for any $\rho \in IR$ such that $U^*(\bar{\theta}; \rho) < 1 - \eta$ there exists an interest rate function $\rho' \in IR$ that generates strictly higher welfare than ρ . Finally, we will conclude by showing that the set of interest rate functions, ρ , satisfying $U^*(\bar{\theta}; \rho) = 1 - \eta$ is non-empty and therefore the optimal interest rate function must satisfy this condition.

To proceed, consider $\rho(U) := m \cdot U + a \in IR$ such that $U^*(\bar{\theta}; \rho) < 1 - \eta$. Next, denote by $\bar{x}$ the value of x that solves

$$1 - \eta = \bar{\theta} \cdot \frac{1 - F_b(\bar{x} + m\eta q + \zeta \cdot q(1 - q)\eta^2 m^2)}{F_l(((1 - \eta) \cdot (1 - q) + q) \cdot \bar{x} + m\eta q)}.$$

Note that $\bar{x}$ is constructed so that if it were the case that $\rho(1 - \eta) = \bar{x}$ then it must be the case that $\mathbb{E}[\rho(U)]_{U=1-\eta} = \bar{x} + m\eta q$ and therefore $U^*(\bar{\theta}; \rho) = 1 - \eta$. Therefore, $U^*(\bar{\theta}; \rho) < 1 - \eta$ implies that $\rho(1 - \eta) > \bar{x}$. Next, we will construct ρ' from ρ . In particular, we will choose $\rho'(U) = a + m' \cdot U$ where m' is some value sufficiently close to m that ensures

$$\bar{x} \leq a + m' \cdot (1 - \eta) < \rho(1 - \eta),$$

where the last inequality implies that $m' < m$ and therefore $\rho'(U) < \rho(U)$ for all $U \in [0, 1]$. Finally, note that by the construction of ρ' we know that there exists $\delta > 0$ such that whenever $m' < m$ but $m - m' < \delta$ then $U^*(\bar{\theta}; \rho') \leq 1 - \eta$ and therefore $\rho' \in IR$. We proceed with the assumption that $m - m' < \delta$ and therefore $\rho' \in IR$.

In what follows, we will use the notation that $\mathbb{E}[\rho(\theta)] := \mathbb{E}[\rho(U)]|_{U=U^*(\bar{\theta}; \rho)}$, $\mathbb{E}[\rho'(\theta)] := \mathbb{E}[\rho'(U)]|_{U=U^*(\bar{\theta}; \rho')}$, $\mathbb{E}[U(\theta)\rho(\theta)] := \mathbb{E}[U \cdot \rho(U)]|_{U=U^*(\bar{\theta}; \rho)}$, and $\mathbb{E}[U'(\theta)\rho'(\theta)] := \mathbb{E}[U \cdot \rho'(U)]|_{U=U^*(\bar{\theta}; \rho')}$. For the next step, we will prove that whenever $\rho'(U) < \rho(U)$ for all $U \in [0, 1]$ then it must be the case that $\mathbb{E}[\rho'(\theta)] < \mathbb{E}[\rho(\theta)]$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. In order to prove this intermediate result,

suppose instead that there exists θ such that $\mathbb{E}[\rho'(\theta)] \geq \mathbb{E}[\rho(\theta)]$. Then, note that by the construction of ρ' we know that $\mathbb{E}[\rho'(\theta)] \geq \mathbb{E}[\rho(\theta)]$ implies that $m' \cdot U^*(\theta; \rho') \geq m \cdot U^*(\theta; \rho)$ which implies that $VAR^*(\theta; \rho') := \iota \cdot (m' \cdot U^*(\theta; \rho'))^2 \geq \iota \cdot (m \cdot U^*(\theta; \rho))^2 =: VAR^*(\theta; \rho)$ and $U^*(\theta; \rho') > U^*(\theta; \rho)$ (given that $m' < m$). We will show that this produces a contradiction. In particular, using the fact that $\mathbb{E}[\rho'(\theta)] \geq \mathbb{E}[\rho(\theta)]$ and $VAR^*(\theta; \rho') \geq VAR^*(\theta; \rho)$ in addition to the fact that $U^*(\theta; \rho') > U^*(\theta; \rho)$ and $m' \cdot U^*(\theta; \rho') \geq m \cdot U^*(\theta; \rho)$ implies by (5.1) that $\mathbb{E}[U'(\theta) \cdot \rho'(\theta)] > \mathbb{E}[U(\theta) \cdot \rho(\theta)]$ and therefore

$$U^*(\theta; \rho') = \theta \cdot \frac{1 - F_b(\mathbb{E}[\rho'(\theta)] + \zeta \cdot VAR^*(\theta; \rho'))}{F_l(\mathbb{E}[U'(\theta) \cdot \rho'(\theta)])} \leq \theta \cdot \frac{1 - F_b(\mathbb{E}[\rho(\theta)] + \zeta \cdot VAR^*(\theta; \rho))}{F_l(\mathbb{E}[U(\theta) \cdot \rho(\theta)])} = U^*(\theta; \rho),$$

which contradicts the fact that $U^*(\theta; \rho') > U^*(\theta; \rho)$. Hence, we have proven by contradiction that $\mathbb{E}[\rho'(\theta)] < \mathbb{E}[\rho(\theta)]$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$.

Next, we note that $\mathbb{E}[\rho'(\theta)] < \mathbb{E}[\rho(\theta)]$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$ implies that for all $\theta \in [\underline{\theta}, \bar{\theta}]$ it must be the case that $m' \cdot U^*(\theta; \rho') < m \cdot U^*(\theta; \rho)$ and therefore $VAR^*(\theta; \rho') < VAR^*(\theta; \rho)$. We will use this to prove that $U^*(\theta; \rho') > U^*(\theta; \rho)$. In particular, by contradiction assume that $U^*(\theta; \rho') < U^*(\theta; \rho)$ for some $\theta \in [\underline{\theta}, \bar{\theta}]$. In that case, we know that $m' \cdot U^*(\theta; \rho') < m \cdot U^*(\theta; \rho)$ and therefore by (5.1) that $\mathbb{E}[U(\theta) \cdot \rho(\theta)] > \mathbb{E}[U'(\theta) \cdot \rho'(\theta)]$. Further, when this is the case then

$$U^*(\theta; \rho) = \theta \cdot \frac{1 - F_b(\mathbb{E}[\rho(\theta)] + \zeta \cdot VAR^*(\theta; \rho))}{F_l(\mathbb{E}[U(\theta) \cdot \rho(\theta)])} < \theta \cdot \frac{1 - F_b(\mathbb{E}[\rho'(\theta)] + \zeta \cdot VAR^*(\theta; \rho'))}{F_l(\mathbb{E}[U'(\theta) \cdot \rho'(\theta)])} = U^*(\theta; \rho'),$$

and therefore $U^*(\theta; \rho') > U^*(\theta; \rho)$, a contradiction given that we have assumed that $U^*(\theta; \rho') < U^*(\theta; \rho)$. Hence, it must be the case that $U^*(\theta; \rho') > U^*(\theta; \rho)$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$.

Finally, in order to prove that ρ' generates higher expected welfare than ρ we will show that welfare under ρ' is strictly higher than welfare under ρ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. In particular, using the additional notation that $U^* = U^*(\theta; \rho)$, $U'^* := U^*(\theta; \rho')$, $\mathbb{E}[\hat{\rho}(\theta)] = \mathbb{E}[\rho^*(\theta)] + \zeta \cdot$

$VAR^*(\theta; \rho)$, and $\mathbb{E}[\hat{\rho}'(\theta)] := \mathbb{E}[\rho'(\theta)] + \zeta \cdot VAR^*(\theta; \rho')$ then note that $\mathcal{W}^{DeFi}(\theta; \rho)$, can be written as the sum of expected surplus as follows¹⁵

$$\begin{aligned}\mathcal{W}^{DeFi}(\theta; \rho) &= \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \mathbb{E}[\hat{\rho}(\theta)]) dF_b(r) + \lambda_t \cdot \int_0^{\mathbb{E}[U(\theta) \cdot \rho(\theta)]} (\mathbb{E}[U(\theta) \cdot \rho(\theta)] - r) dF_l(r) = \\ &\mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho)) dF_b(r) - \lambda_t \int_0^{\mathbb{E}[U(\theta) \rho(\theta)]} r dF_l(r),\end{aligned}$$

where the last equality uses the fact that the interest paid by borrowers is equal to the interest payments received by the lenders when there is no withdrawal shock and, the the fact that we have assumed that the shocked lenders that withdraw receive the same utility (and therefore generate the same welfare) as the non-shocked lenders. Therefore, from the welfare perspective $\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho(\theta)] = \lambda_t F_l(U(\theta)\rho(\theta))\mathbb{E}[U(\theta)\rho(\theta)]$. Using the same logic we can see that

$$\mathcal{W}^{DeFi}(\theta; \rho') = \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}'(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho')) dF_b(r) - \lambda_t \int_0^{\mathbb{E}[U'(\theta)\rho'(\theta)]} r dF_l(r).$$

Therefore,

$$\begin{aligned}\mathcal{W}^{DeFi}(\theta; \rho) - \mathcal{W}^{DeFi}(\theta; \rho') &= \\ \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho)) dF_b(r) &- \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}'(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho')) dF_b(r) + \lambda_t \int_{\mathbb{E}[U^* \rho(\theta)]}^{\mathbb{E}[U'(\theta)\rho'(\theta)]} r dF_l(r).\end{aligned}$$

Next, note that $VAR^*(\theta, \rho) > VAR^*(\theta, \rho')$ and $\mathbb{E}[\hat{\rho}(\theta)] > \mathbb{E}[\hat{\rho}'(\theta)]$ implies that

$$\mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho)) dF_b(r) < \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}'(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho')) dF_b,$$

and therefore if $\mathbb{E}[U^* \rho^*] \geq \mathbb{E}[U'^* \rho'^*]$ then it must be the case that $\mathcal{W}^{DeFi}(\theta; \rho) - \mathcal{W}^{DeFi}(\theta; \rho') < 0$ and we have proven our result. Thus, in what follows we will proceed with the assumption

¹⁵Namely, the following expression is equal to the sum of all borrower and lender pay-offs minus a constant (equal to $\lambda_t \int_0^{+\infty} r dF_l(r)$).

that $\mathbb{E}[U^*\rho^*] < \mathbb{E}[U'^*\rho'^*]$.

For the next step, note that $VAR^*(\theta, \rho) > VAR^*(\theta, \rho')$ implies that

$$\mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho)) dF_b(r) < \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \zeta \cdot VAR^*(\theta; \rho')) dF_b(r),$$

and therefore

$$\begin{aligned} \mathcal{W}^{DeFi}(\theta; \rho) - \mathcal{W}^{DeFi}(\theta; \rho') &< \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{\mathbb{E}[\hat{\rho}'(\theta)]} (r - \zeta \cdot VAR^*(\theta; \rho')) dF_b(r) + \lambda_t \int_{\mathbb{E}[U(\theta)\rho(\theta)]}^{\mathbb{E}[U'(\theta)\rho'(\theta)]} r dF_l(r) \leq \\ &\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho'(\theta)] - \mu_t(1 - F_b(\mathbb{E}[\hat{\rho}'(\theta)]))\mathbb{E}[\rho'(\theta)] + \\ &\lambda_t(F_l(\mathbb{E}[U'(\theta)\rho'(\theta)])\mathbb{E}[U'(\theta)\rho'(\theta)] - \lambda_t(F_l(\mathbb{E}[U(\theta)\rho(\theta)])\mathbb{E}[U'(\theta)\rho'(\theta)] = \\ &\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho'(\theta)] - \lambda_t F_l(\mathbb{E}[U(\theta)\rho(\theta)])\mathbb{E}[U'(\theta)\rho'(\theta)]. \end{aligned}$$

Where the previous equality again uses the fact that $\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}'(\theta)]))\mathbb{E}[\rho'(\theta)] = \lambda_t F_l(\mathbb{E}[U'(\theta)\rho'(\theta)])\mathbb{E}[U'(\theta)\rho'(\theta)]$.

Next, using the fact that we are in the case whereby $\mathbb{E}[U(\theta)\rho(\theta)] < \mathbb{E}[U'(\theta)\rho'(\theta)]$ and noting again that $\lambda_t F_l(\mathbb{E}[U(\theta)\rho(\theta)])\mathbb{E}[U(\theta)\rho(\theta)] = \mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho(\theta)]$ we can see that

$$\begin{aligned} &\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho'(\theta)] - \lambda_t F_l(\mathbb{E}[U(\theta)\rho(\theta)])\mathbb{E}[U'(\theta)\rho'(\theta)] < \\ &\mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))\mathbb{E}[\rho'(\theta)] - \lambda_t F_l(\mathbb{E}[U(\theta)\rho(\theta)])\mathbb{E}[U(\theta)\rho(\theta)] \\ &= \mu_t(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))(\mathbb{E}[\rho'(\theta)] - \mathbb{E}[\rho(\theta)]) < 0. \end{aligned}$$

where the final inequality comes from the fact that we have proven above that $\mathbb{E}[\rho'(\theta)] < \mathbb{E}[\rho(\theta)]$. Hence, we have proven that in all cases $\mathcal{W}^{DeFi}(\theta; \rho) - \mathcal{W}^{DeFi}(\theta; \rho') < 0$.

We have shown that for any interest rate function $\rho \in IR$ such that $U^*(\theta; \rho) < 1 - \eta$ there exists another interest rate function $\rho' \in IR$ that generates strictly higher welfare than ρ for

all $\theta \in [\underline{\theta}, \bar{\theta}]$. Therefore, the optimal interest rate function ρ must satisfy $U^*(\bar{\theta}, \rho) = 1 - \eta$. Finally, note that we prove that the set of functions $\rho \in IR$ such that $U^*(\bar{\theta}, \rho) = 1 - \eta$ is non-empty in the proof of Proposition 5.2 and therefore we preclude this step here.

□

A.11 Proof of Proposition 5.2

Proof. In order to prove the result, first note that for any m the solution to (5.3) must exist. In particular, fixing m , the numerator of (5.3) is strictly decreasing in x while the denominator is strictly increasing in x . Further, as $x \rightarrow +\infty$ then the right hand side of (5.3) approaches 0. Further, let

$$\hat{x}(m) := -\frac{q(1-q)\eta^2}{(1-q)(1-\eta) + q} \cdot m,$$

then it can be seen that for any m , as $x \rightarrow \hat{x}(m)$ then $\mathbb{E}[U\rho(U)]|_{U=1-\eta} \rightarrow 0$. Therefore, whenever $x \rightarrow^+ \hat{x}(m)$ then the right hand side of (5.3) approaches $+\infty$. Thus, for any $m \in [0, +\infty)$ there must exist a unique solution to (5.3) which we denote by $\bar{x}(m)$. Finally, note that by construction it must be the case that $\bar{x}(m) > \hat{x}(m)$ and therefore $\mathbb{E}[U\rho(U)]|_{U=1-\eta} > 0$ for all $m \in [0, +\infty)$ when setting $\mathbb{E}[\rho(U)]|_{U=1-\eta} = \bar{x}(m)$. In addition, note that for any x , increasing m decreases the numerator of (5.3) and increases the denominator of (5.3), thus it must be the case that $\frac{d}{dm}\bar{x}(m) < 0$. In addition, note that as $m \rightarrow +\infty$ it must be the case that $\bar{x}(m) \rightarrow -\infty$.

Next note that when $\mathbb{E}[\rho(U)]|_{U=1-\eta} = \bar{x}(m)$ then the term in the numerator of (5.3) is $x + \iota \cdot (m(1-\eta))^2 = \mathbb{E}[\rho(1-\eta)] + \zeta \cdot VAR(1-\eta; \rho) > 0$ where $VAR(U; \rho)$ denotes the variance of interest rates when the utilization rate is U and the PLF interest rate function is ρ . Further, noting that whenever the utilization rate at the start of the period (before the withdrawal shock) is $U = 1 - \eta$, then $U^+ = 1$. Thus, if we denote by $\tilde{x}$ the interest paid in

the state where there is no withdrawal shock, then $\tilde{x} + \eta \cdot m$ must be the interest paid when there is a withdrawal shock. Thus, $\mathbb{E}[\rho(1 - \eta)] = \bar{x}(m)$ implies that $\tilde{x} = \bar{x}(m) - m \cdot q \cdot \eta$. Hence, $\tilde{x} = \bar{x}(m) - m \cdot q \cdot \eta$ implies that the term in the denominator of (5.3) is $(1 - q)(1 - \eta)(x - m \cdot q \cdot \eta) + q(x + m \cdot (1 - q) \cdot \eta) = \mathbb{E}[U\rho(U)]|_{U=1-\eta}$. Hence, if $U^*(\bar{\theta}; \rho) = 1 - \eta$ then it must be the case that

$$1 - \eta = \bar{\theta} \cdot \frac{1 - F_b(\mathbb{E}[\rho(U)] + \zeta \cdot \text{VAR}(U; \rho))}{F_l(\mathbb{E}[U\rho(U)])}|_{U=1-\eta},$$

which by the uniqueness of $\bar{x}(m)$ implies that it must be the case that $\mathbb{E}[\rho(U)]|_{U=U^*(\bar{\theta}; \rho)} = \bar{x}(m)$. $\square$

A.12 Proof of Proposition 5.3

Proof. Let $\rho(U) = a^*(m) + m \cdot U$ for some $m \in [0, +\infty)$. In order to prove existence of an optimal $m^* \in [0, +\infty)$ we only need to show that for all $\theta \in [\underline{\theta}, \bar{\theta}]$ there exists values m' and $\hat{m} > m'$ such that for all $\tilde{m} > \hat{m}$ it is the case that

$$\mathcal{W}^{DeFi}(\theta; \rho)|_{m=m'} > \mathcal{W}^{DeFi}(\theta; \rho)|_{m=\tilde{m}}.$$

Before proceeding, first note that in this case, $U^*(\theta; \rho)$ is the solution to the following equation.

$$U = \theta \cdot \frac{1 - F_b(a^*(m) + \iota \cdot (m \cdot U)^2)}{F_l(U \cdot ((1 - q) \cdot (a^*(m) + m \cdot U) + \frac{q}{1-\eta} \cdot (a^*(m) + m \cdot \frac{U}{1-\eta}))},$$

and therefore given that the right hand side of this equation is strictly decreasing in U then it must be the case that $U^*(\theta; \rho)|_{m=m'} < U^*(\bar{\theta}; \rho)|_{m=m'}$ for any $\theta \in [\underline{\theta}, \bar{\theta})$ and any $m' \in [0, +\infty)$.

In what follows, recall that $\mathbb{E}[\hat{\rho}(\theta)] := \mathbb{E}[\hat{\rho}(U)]|_{U=U^*(\theta; \rho)}$ and that $\mathbb{E}[U(\theta)\rho(\theta)] := \mathbb{E}[U \cdot$

$\rho(U)]|_{U=U^*(\theta;\rho)}$. Next, note that for all $m \in [0, +\infty)$ and all $\theta \in [\underline{\theta}, \bar{\theta}]$ it is the case that

$$\mathbb{E}[U(\theta)\rho(\theta)]|_{m=m'} \leq \mathbb{E}[U(\bar{\theta})\rho(\bar{\theta})]|_{m=m'}, \quad (\text{A.10})$$

coming from the fact that $\mathbb{E}[U\rho(U)]|_{m=m'}$ is strictly decreasing in U and $U^*(\theta;\rho)|_{m=m'} < U^*(\bar{\theta};\rho)|_{m=m'}$.

Next, we will show that $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] = 0$. Namely, we know from the proof of Proposition 5.2 that when setting $a = a^*(m)$ then we ensure that $\mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] > 0$ for all $m \in [0, +\infty)$ and therefore it must be the case that $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] \geq 0$. Next, suppose by contradiction that $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] > 0$. In this case, there must exist a constant B such that $\lim_{m \rightarrow +\infty} \bar{x}'(m) > B$. Otherwise, $\lim_{m \rightarrow +\infty} \bar{x}'(m) \rightarrow -\infty$ and therefore

$$\lim_{m \rightarrow +\infty} \frac{d}{dm} \mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] = \lim_{m \rightarrow +\infty} ((1-\eta)(1-q) + q) \cdot \bar{x}'(m) + q(1-q)\eta^2 \rightarrow -\infty,$$

contradicting the fact that $\lim_{m \rightarrow +\infty} \mathbb{E}[U\rho(U)]|_{U=1-\eta} > 0$. Next, note that $\lim_{m \rightarrow +\infty} \bar{x}'(m) > B$ for some constant B implies that

$$\lim_{m \rightarrow +\infty} \frac{d}{dm} \mathbb{E}[\hat{\rho}(\theta)] = \lim_{m \rightarrow +\infty} \bar{x}'(m) + 2m \cdot \zeta_\iota(1-\eta)^2 = +\infty.$$

Yet, this cannot be true as in that case $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\bar{\theta}) \cdot \rho(\bar{\theta})] > 0$ while $\lim_{m \rightarrow +\infty} \mathbb{E}[\hat{\rho}(U)]|_{U=1-\eta} = +\infty$ (by the continuity of $\mathbb{E}[\hat{\rho}(U)]$) and therefore $\lim_{m \rightarrow +\infty} U^*(\bar{\theta};\rho) = 0$ contradicting the fact that $U^*(\bar{\theta};\rho) = 1-\eta$ for all $m \in [0, +\infty)$ by construction.

Hence, noting that $\mathbb{E}[U(\theta)\rho(\theta)] \leq \mathbb{E}[U(\bar{\theta})\rho(\bar{\theta})]$ for all $m \in [0, +\infty)$ and $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\bar{\theta})\rho(\bar{\theta})] = 0$ then it must be the case that $\lim_{m \rightarrow +\infty} \mathbb{E}[U(\theta)\rho(\theta)] = 0$. Thus, fixing $m' \in [0, +\infty)$ then for all $\theta \in [\underline{\theta}, \bar{\theta}]$ there must exist $\hat{m} \in (m', +\infty)$ such that for all $\tilde{m} > \hat{m}$

$$\mathbb{E}[U(\theta)\rho(\theta)]|_{m=m'} > \mathbb{E}[U(\theta)\rho(\theta)]|_{m=\tilde{m}}.$$

We will show that this implies that there must exist $\hat{m}' \in (\hat{m}, +\infty)$ such that for all $\tilde{m} > \hat{m}'$ it must be the case that

$$\mathbb{E}[\hat{\rho}(\theta)]|_{m=\tilde{m}} > \mathbb{E}[\hat{\rho}(\theta)]|_{m=m'}.$$

To prove this result, suppose by contradiction that for all $\tilde{m} > \hat{m}$ it is the case that

$$\mathbb{E}[\hat{\rho}(\theta)]|_{m=\tilde{m}} < \mathbb{E}[\hat{\rho}(\theta)]|_{m=m'}.$$

Then, we know that for all $\tilde{m} \in (\hat{m}, +\infty)$

$$U^\star(\theta; \rho)|_{m=\tilde{m}} = \theta \cdot \frac{(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))}{F_l(\mathbb{E}[U(\theta)\rho(\theta)])} \Big|_{m=\tilde{m}} > \theta \cdot \frac{(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))|_{m=m'}}{F_l(\mathbb{E}[U(\theta)\rho(\theta)])|_{m=\tilde{m}}}.$$

Yet, from the above result we know that

$$\lim_{\tilde{m} \rightarrow +\infty} \theta \cdot \frac{(1 - F_b(\mathbb{E}[\hat{\rho}(\theta)]))|_{m'}}{F_l(\mathbb{E}[U(\theta)\rho(\theta)])|_{\tilde{m}}} = +\infty,$$

contradicting the fact that $U^\star(\theta; \rho) \leq U^\star(\bar{\theta}; \rho) = 1 - \eta$ for all $m \in [0, +\infty)$.

In conclusion, we have shown that for any $m' \in [0, +\infty)$ there exists $\hat{m}'$ such that for all $\tilde{m} > \hat{m}'$ we know that $\mathbb{E}[U(\theta)\rho(\theta)]|_{m=m'} > \mathbb{E}[U(\theta)\rho(\theta)]|_{m=\tilde{m}}$ and $\mathbb{E}[\hat{\rho}(\theta)]|_{m=m'} < \mathbb{E}[\hat{\rho}(\theta)]|_{m=\tilde{m}}$ and therefore noting that

$$\mathcal{W}^{DeFi}(\theta; \rho) = \mu_t \cdot \int_{\mathbb{E}[\hat{\rho}(\theta)]}^{+\infty} (r - \mathbb{E}[\hat{\rho}(\theta)]) dF_b(r) + \lambda_t \cdot \int_0^{\mathbb{E}[U(\theta) \cdot \rho(\theta)]} (\mathbb{E}[U(\theta) \cdot \rho(\theta)] - r) dF_l(r),$$

for any $m \in [0, +\infty)$ then it must be the case that

$$\mathcal{W}^{DeFi}(\theta; \rho)|_{m=\tilde{m}} < \mathcal{W}^{DeFi}(\theta; \rho)|_{m=m'},$$

for all $\tilde{m} > \hat{m}'$ and therefore $m^\star(\theta) \leq \hat{m}' < +\infty$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. □